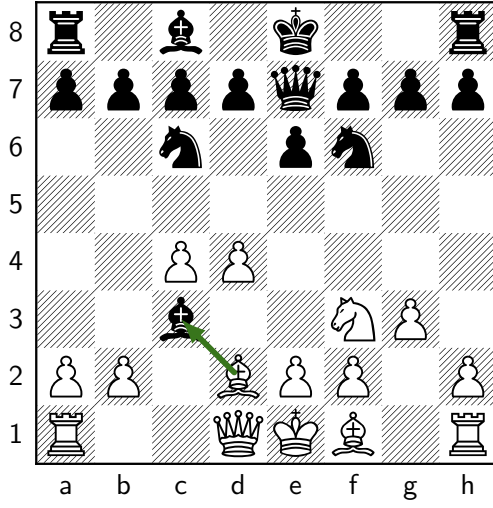


# Melanie-XRAY — Bogo-Indian

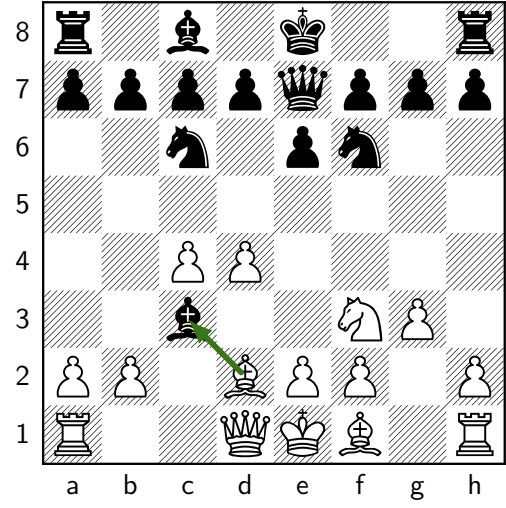
Melanie (Deep\_Vision)

## Abstract

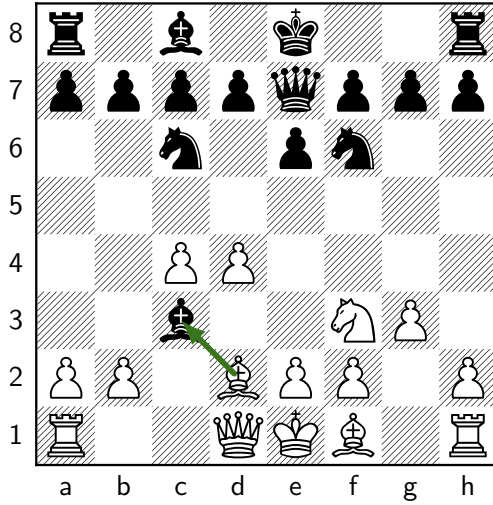
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie's move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under  $\mu$  – cross-type maps (a rook playing a queen's role) are called out, as no material/eval engine explains a move that way.



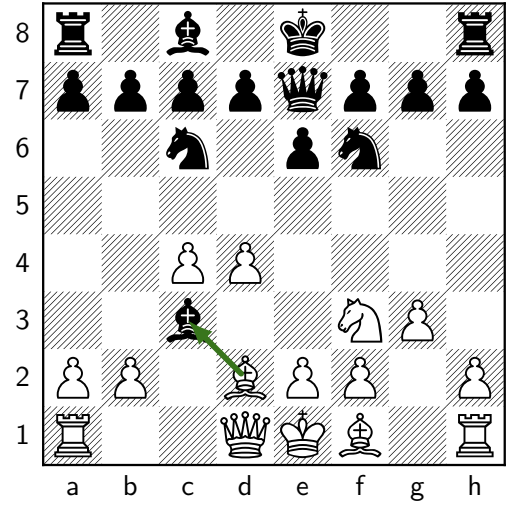
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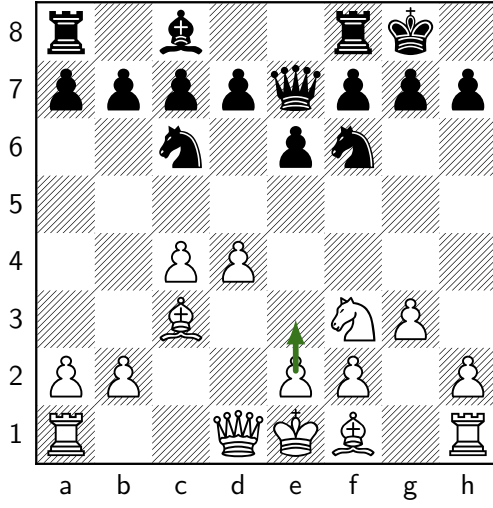


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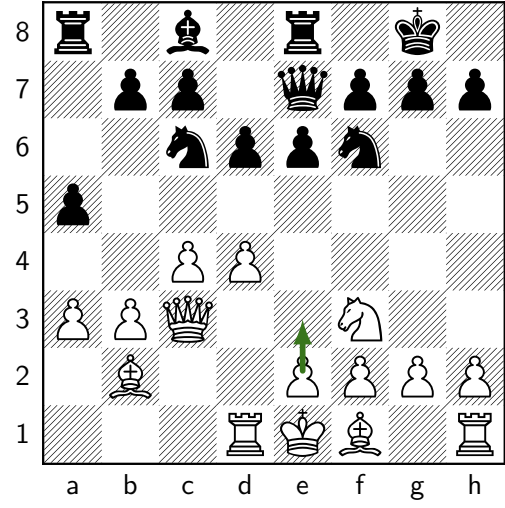


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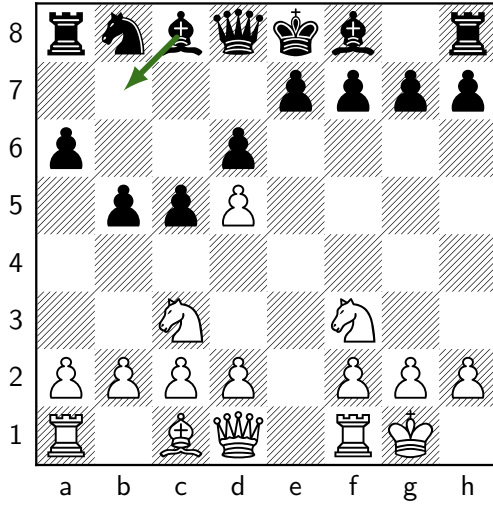
Figure 1: **Move 7, White: Bxc3.** (a) Our position; Melanie plays Bxc3 (d2c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3642` (ply 12) [structural distance  $d=0$ , QAP obj=210]: the master played Bxc3; under  $\mu$  our Bd2 plays the role of that game’s Bd2; cross-type roles: our rh8 as their nf6, our nf6 as their rh8; transferred move d2c3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3643` (ply 12) [structural distance  $d=0$ , QAP obj=210]: the master played Bxc3; under  $\mu$  our Bd2 plays the role of that game’s Bd2; cross-type roles: our rh8 as their nf6, our nf6 as their rh8; transferred move d2c3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1512` (ply 12) [structural distance  $d=0$ , QAP obj=210]: the master played Bxc3; under  $\mu$  our Bd2 plays the role of that game’s Bd2; cross-type roles: our rh8 as their nf6, our nf6 as their rh8; transferred move d2c3 (**endorses this move**). Synthesis: 47 retrieved analogs transferred to Bxc3 (summed QAP objective 7356, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



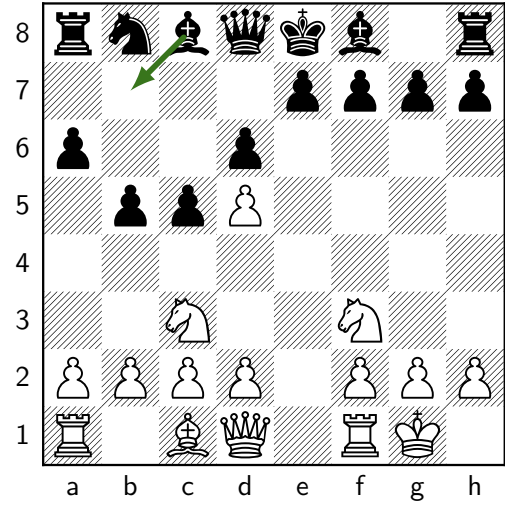
(a)



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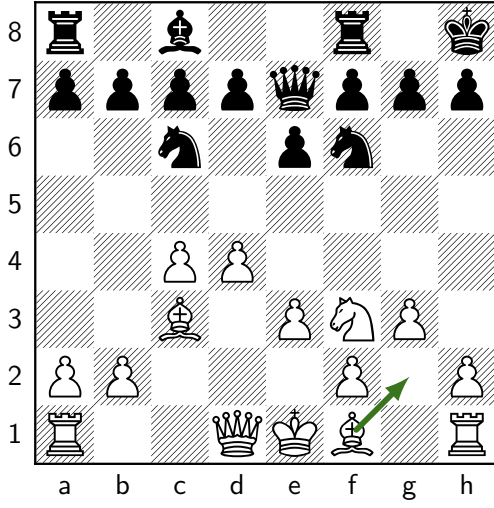


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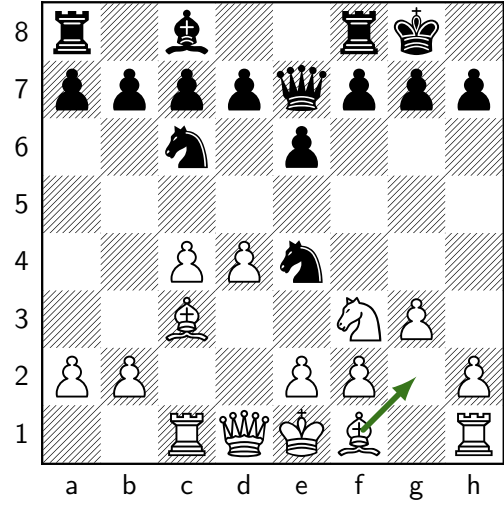


(d)

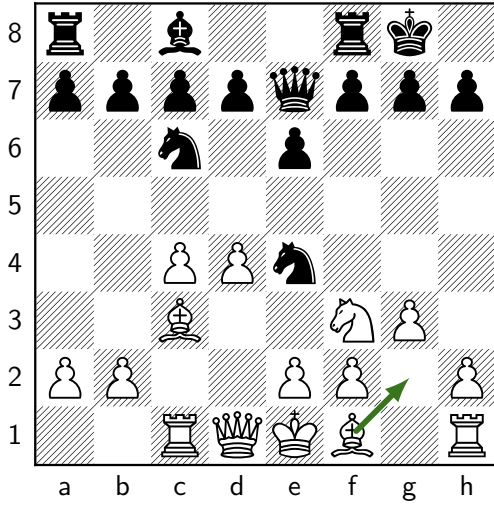
Figure 2: **Move 8, White: e3**. (a) Our position; Melanie plays e3 (e2e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#856` (ply 20) [structural distance  $d=50$ , QAP obj=43]: the master played e3; under  $\mu$  our  $Pe2$  plays the role of that game’s  $Pe2$ ; cross-type roles: our  $Qd1$  as their  $Rd1$ , our  $pb7$  as their  $Bb2$ ; transferred move e2e3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#206` (ply 13) [structural distance  $d=48$ , QAP obj=33]: the master played Bb7; under  $\mu$  our  $Pe2$  plays the role of that game’s  $Bc1$  (a **cross-type** correspondence); cross-type roles: our  $pd7$  as their  $Nb8$ , our  $Pf2$  as their  $Bf8$ ; transferred move e2e3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#207` (ply 13) [structural distance  $d=48$ , QAP obj=33]: the master played Bb7; under  $\mu$  our  $Pe2$  plays the role of that game’s  $Bc1$  (a **cross-type** correspondence); cross-type roles: our  $pd7$  as their  $Nb8$ , our  $Pf2$  as their  $Bf8$ ; transferred move e2e3 (**endorses this move**). Synthesis: 17 retrieved analogs transferred to e3 (summed QAP objective 430, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



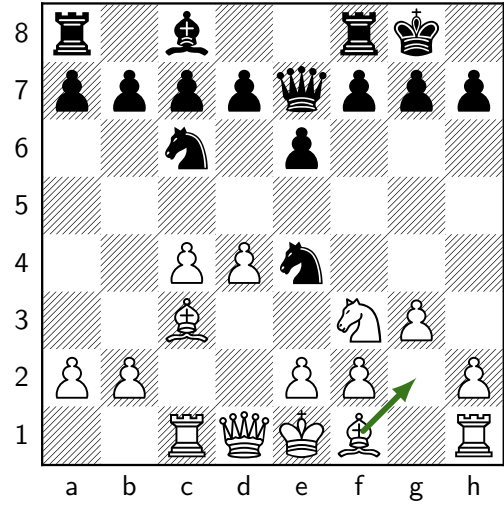
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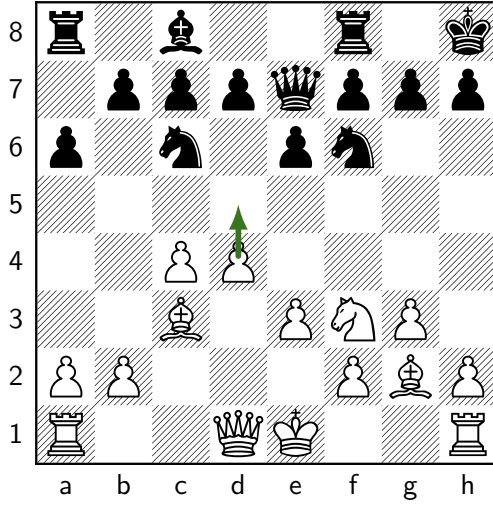


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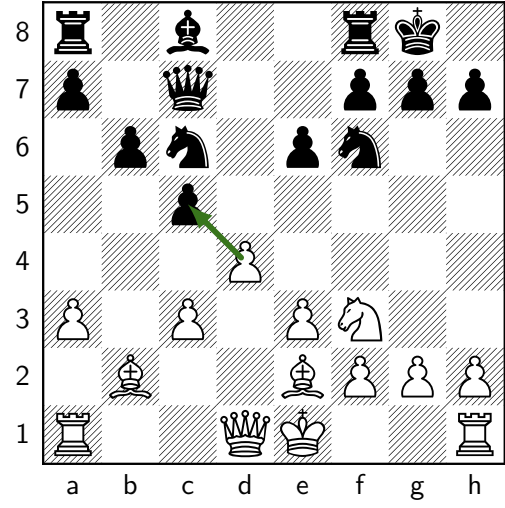


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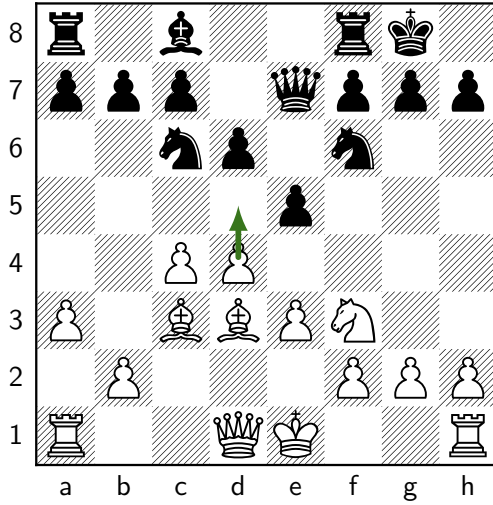
Figure 3: **Move 9, White: Bg2.** (a) Our position; Melanie plays Bg2 (f1g2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3642` (ply 16) [structural distance  $d=65$ , QAP obj=26]: the master played Bg2; under  $\mu$  our Bf1 plays the role of that game’s Bf1; cross-type roles: our Bc3 as their Qd1, our Nf3 as their Ph2; transferred move f1g2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3643` (ply 16) [structural distance  $d=65$ , QAP obj=26]: the master played Bg2; under  $\mu$  our Bf1 plays the role of that game’s Bf1; cross-type roles: our Bc3 as their Qd1, our Nf3 as their Ph2; transferred move f1g2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1512` (ply 16) [structural distance  $d=65$ , QAP obj=26]: the master played Bg2; under  $\mu$  our Bf1 plays the role of that game’s Bf1; cross-type roles: our Bc3 as their Qd1, our Nf3 as their Ph2; transferred move f1g2 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Bg2 (summed QAP objective 104, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



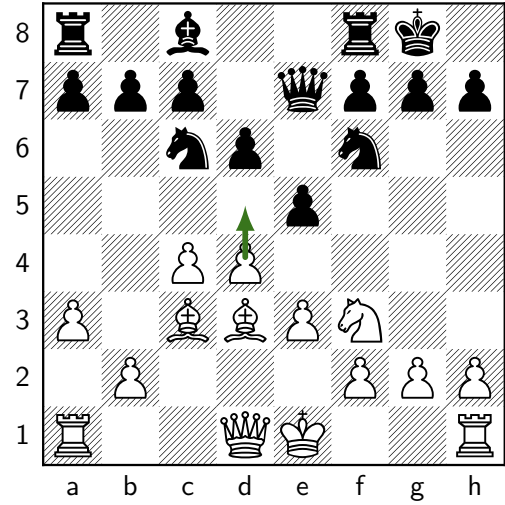
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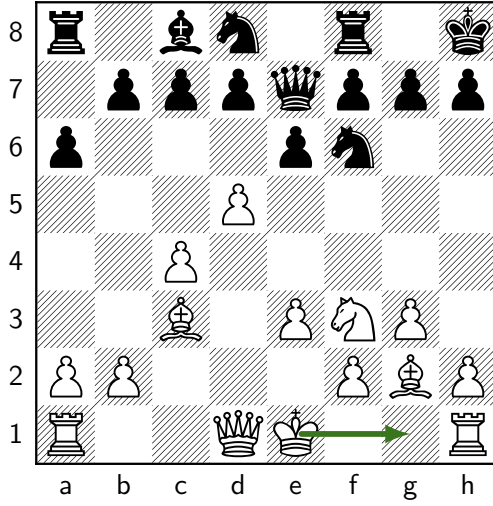


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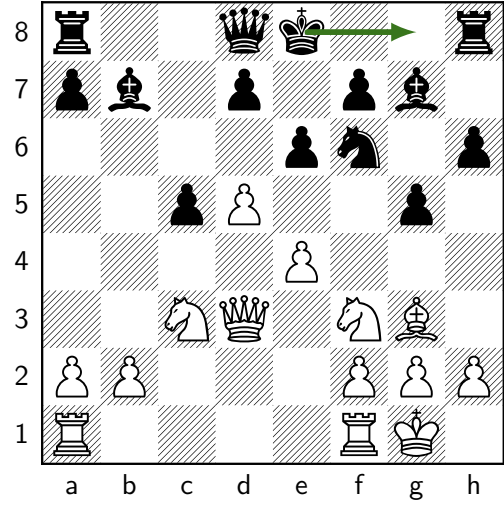


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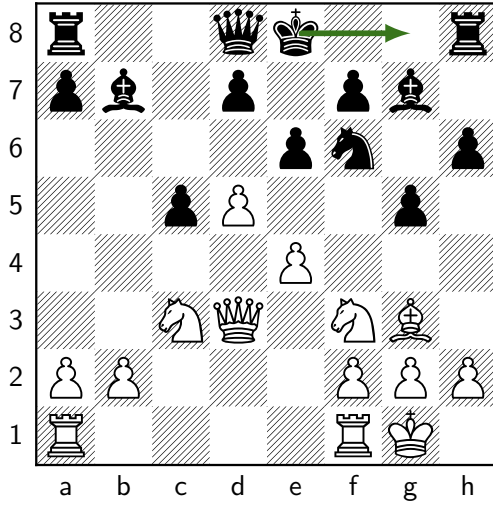
Figure 4: **Move 10, White: d5**. (a) Our position; Melanie plays d5 (d4d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#68` (ply 22) [structural distance  $d=74$ , QAP obj=34]: the master played `dx5`; under  $\mu$  our  $Pd4$  plays the role of that game’s  $Pd4$ ; cross-type roles: our  $pa6$  as their  $Be2$ , our  $bc8$  as their  $Ph2$ ; transferred move `d4d5` (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3433` (ply 18) [structural distance  $d=78$ , QAP obj=33]: the master played `d5`; under  $\mu$  our  $Pd4$  plays the role of that game’s  $Pd4$ ; cross-type roles: our  $Bg2$  as their  $Pf2$ , our  $Nf3$  as their  $Pa3$ ; transferred move `d4d5` (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3434` (ply 18) [structural distance  $d=78$ , QAP obj=33]: the master played `d5`; under  $\mu$  our  $Pd4$  plays the role of that game’s  $Pd4$ ; cross-type roles: our  $Bg2$  as their  $Pf2$ , our  $Nf3$  as their  $Pa3$ ; transferred move `d4d5` (**endorses this move**). Synthesis: 17 retrieved analogs transferred to d5 (summed QAP objective 313, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



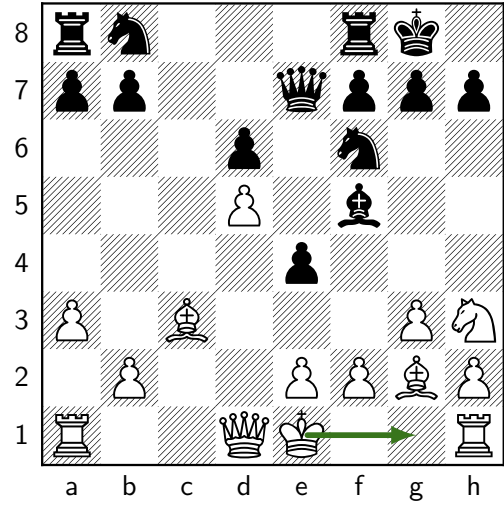
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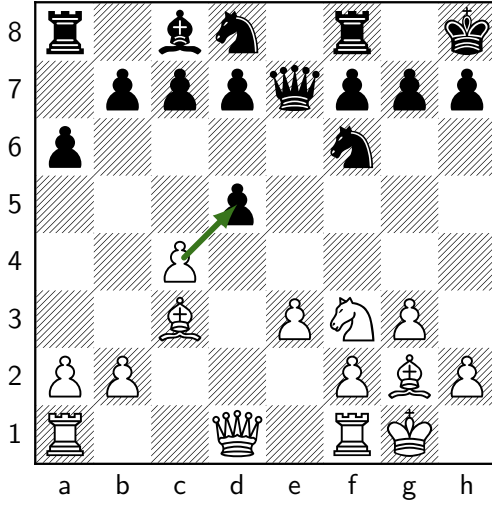


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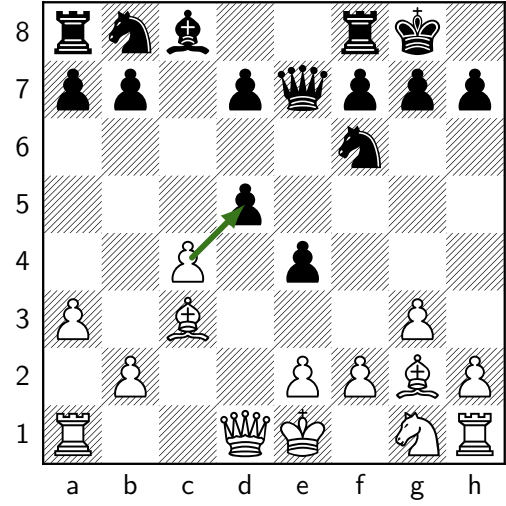


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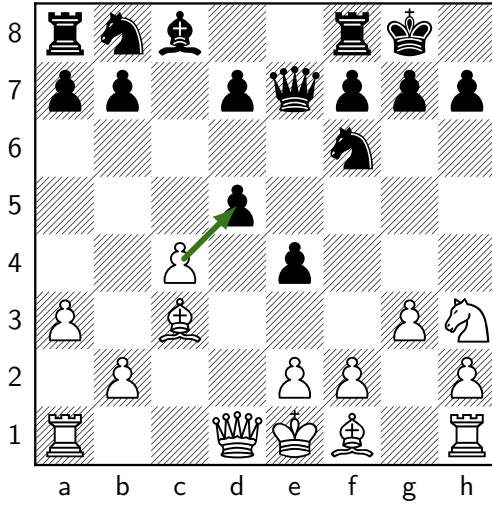
Figure 5: **Move 11, White: 0-0.** (a) Our position; Melanie plays 0-0 (**e1g1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#694` (ply 27) [structural distance  $d=99$ , QAP obj=115]: the master played 0-0; under  $\mu$  our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *nf6* as their *Bb7*, our *nd8* as their *Bg7*; transferred move **e1g1** (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#695` (ply 27) [structural distance  $d=99$ , QAP obj=115]: the master played 0-0; under  $\mu$  our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *nf6* as their *Bb7*, our *nd8* as their *Bg7*; transferred move **e1g1** (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#442` (ply 22) [structural distance  $d=95$ , QAP obj=68]: the master played 0-0; under  $\mu$  our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *bc8* as their *Rh1*, our *pe6* as their *Nh3*; transferred move **e1g1** (**endorses this move**). Synthesis: 28 retrieved analogs transferred to 0-0 (summed QAP objective 1321, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



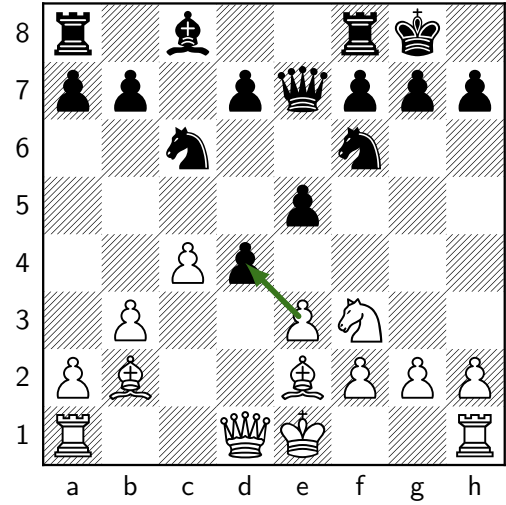
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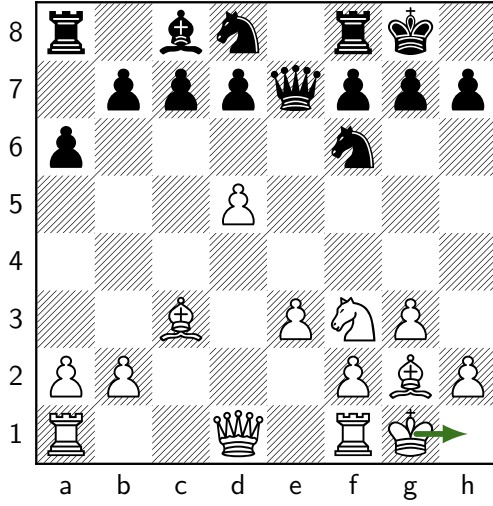


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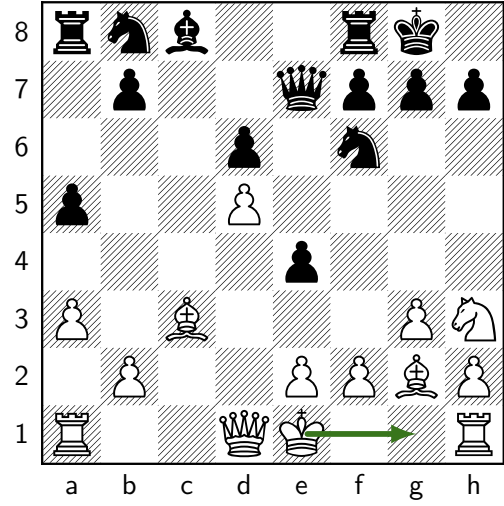


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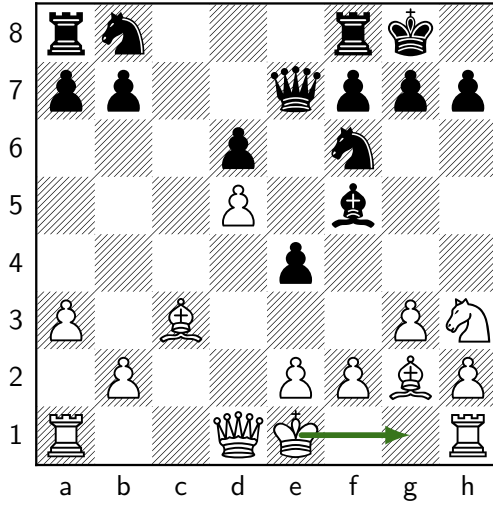
Figure 6: **Move 12, White: cxd5.** (a) Our position; Melanie plays cxd5 (c4d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0003#441 (ply 18) [structural distance  $d=55$ , QAP obj=92]: the master played cxd5; under  $\mu$  our  $Pc4$  plays the role of that game’s  $Pc4$ ; cross-type roles: our  $pa6$  as their  $Ng1$ , our  $Pb2$  as their  $Rh1$ ; transferred move c4d5 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0003#442 (ply 18) [structural distance  $d=63$ , QAP obj=87]: the master played cxd5; under  $\mu$  our  $Pc4$  plays the role of that game’s  $Pc4$ ; cross-type roles: our  $Pg3$  as their  $Ra1$ , our  $Nf3$  as their  $Rh1$ ; transferred move c4d5 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0001#1424 (ply 20) [structural distance  $d=82$ , QAP obj=81]: the master played exd4; under  $\mu$  our  $Pc4$  plays the role of that game’s  $Pe3$ ; cross-type roles: our  $Pc4$  as their  $Bb2$ , our  $Nf3$  as their  $Be2$ ; transferred move c4d5 (**endorses this move**). Synthesis: 31 retrieved analogs transferred to cxd5 (summed QAP objective 1634, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



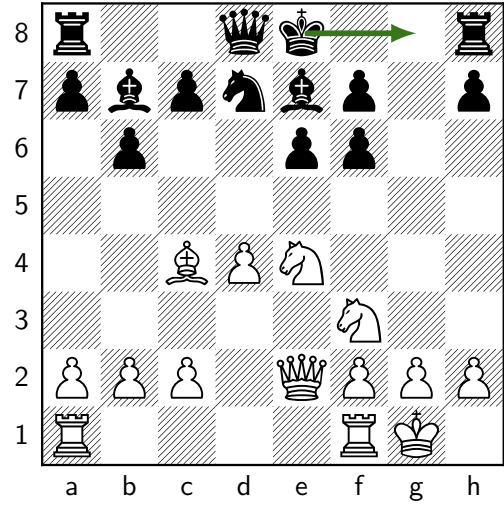
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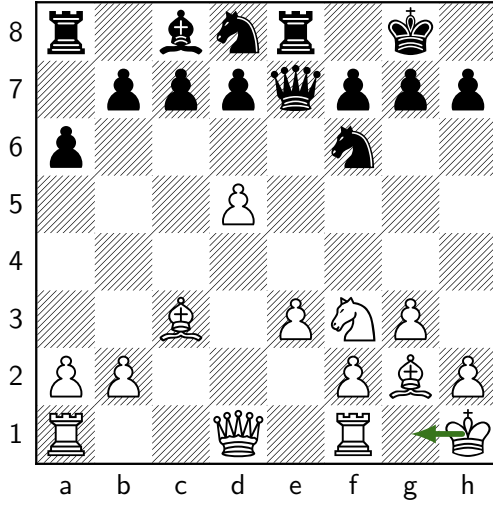
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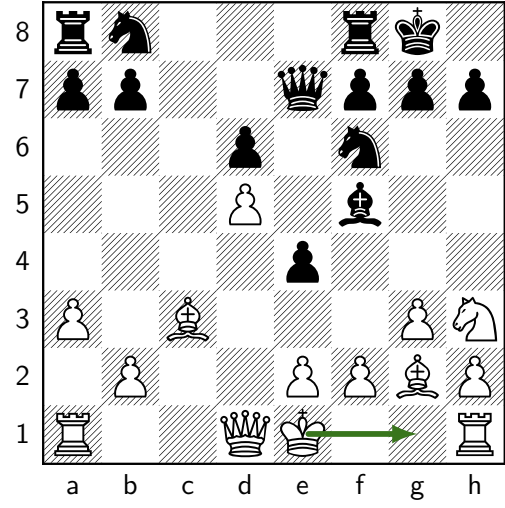
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Figure 7: **Move 13, White: Kh1.** (a) Our position; Melanie plays Kh1 (**g1h1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#441` (ply 22) [structural distance  $d=75$ , QAP obj=76]: the master played 0-0; under  $\mu$  our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Qd1* as their *Ra1*, our *Rf1* as their *Qd1*; transferred move **g1h1 (endorses this move)**. (c) Retrieved from `ccr1_top_p0003#442` (ply 22) [structural distance  $d=54$ , QAP obj=65]: the master played 0-0; under  $\mu$  our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Qd1* as their *Ra1*, our *Rf1* as their *Qd1*; transferred move **g1h1 (endorses this move)**. (d) Retrieved from `ccr1_top_p0000#196` (ply 19) [structural distance  $d=70$ , QAP obj=51]: the master played 0-0; under  $\mu$  our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Qd1* as their *Ra8*, our *Rf1* as their *Qd8*; transferred move **g1h1 (endorses this move)**. Synthesis: 4 retrieved analogs transferred to Kh1 (summed QAP objective 243, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

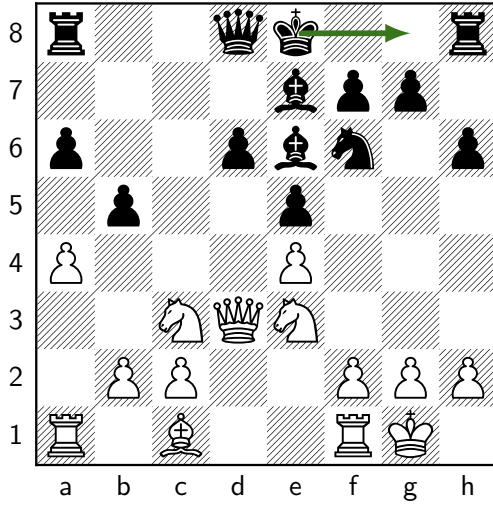




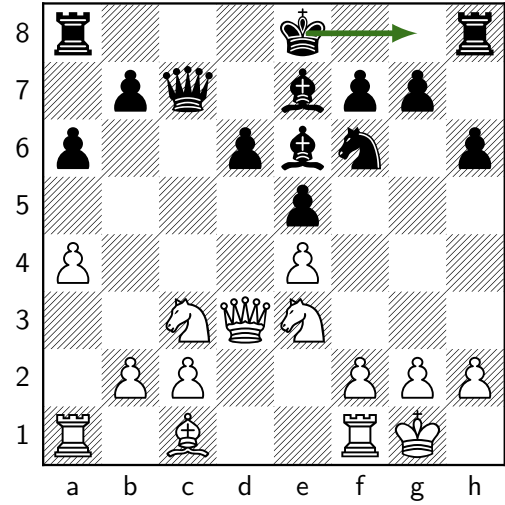
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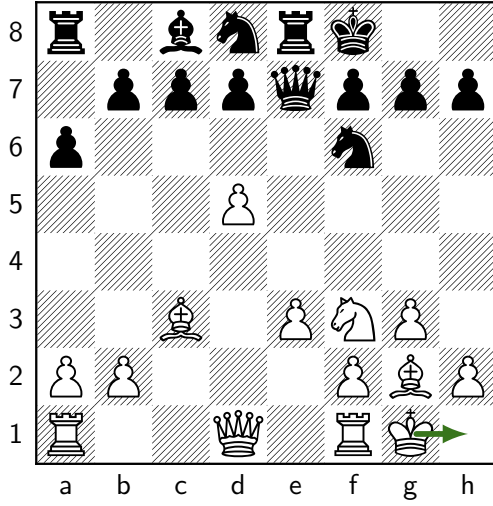


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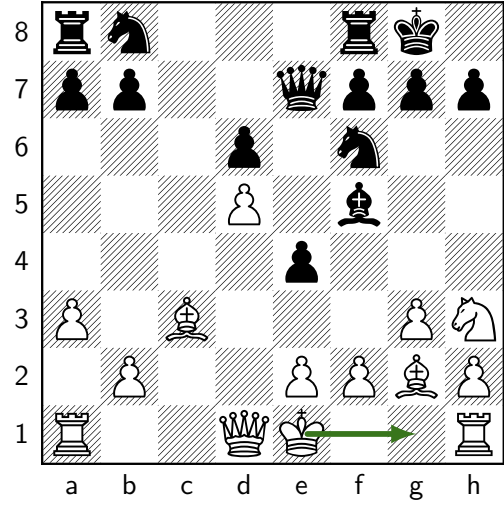


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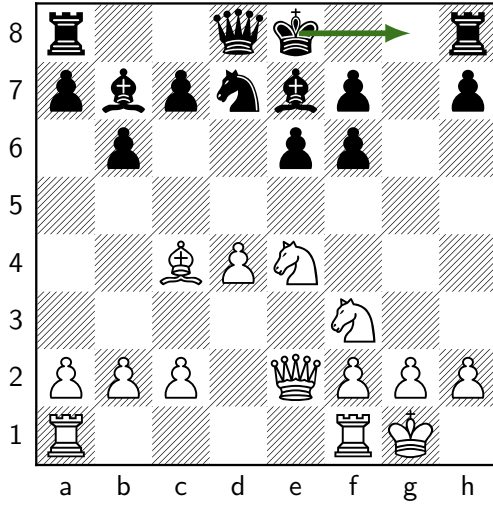
Figure 8: **Move 14, White: Kg1.** (a) Our position; Melanie plays Kg1 (**h1g1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#442` (ply 22) [structural distance  $d=64$ , QAP obj=54]: the master played 0-0; under  $\mu$  our *Kh1* plays the role of that game’s *Ke1*; cross-type roles: our *Qd1* as their *Ra1*, our *Bg2* as their *Qd1*; transferred move **h1g1** (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#277` (ply 25) [structural distance  $d=75$ , QAP obj=53]: the master played 0-0; under  $\mu$  our *Kh1* plays the role of that game’s *Ke1*; cross-type roles: our *Qd1* as their *Ra8*, our *Bg2* as their *Qd8*; transferred move **h1g1** (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1280` (ply 25) [structural distance  $d=77$ , QAP obj=29]: the master played 0-0; under  $\mu$  our *Kh1* plays the role of that game’s *Ke1*; cross-type roles: our *Bg2* as their *Ra8*, our *Bc3* as their *Qc7*; transferred move **h1g1** (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Kg1 (summed QAP objective 222, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



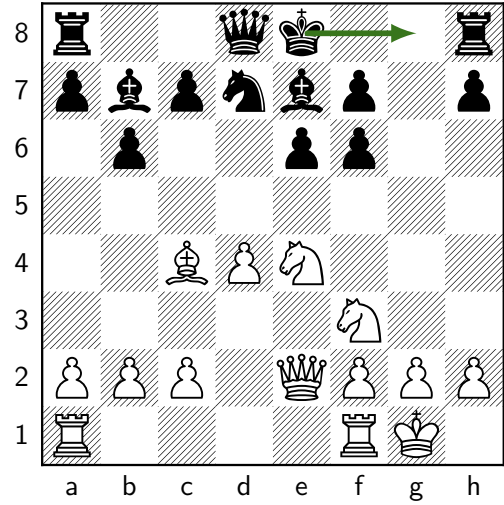
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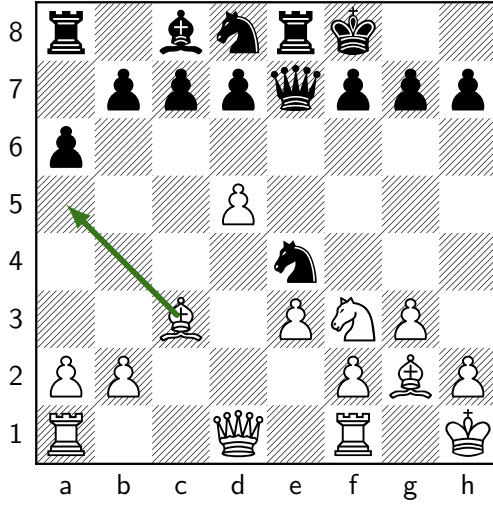


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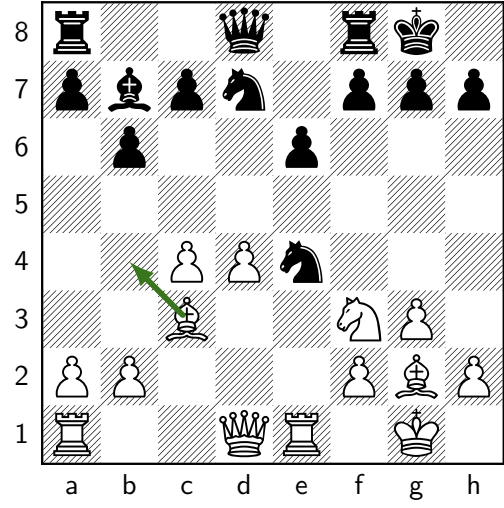


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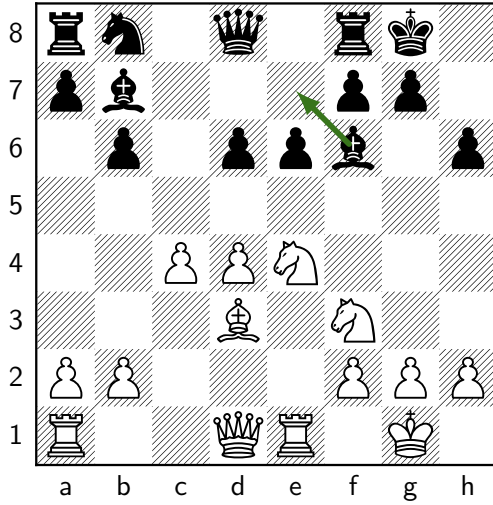
Figure 9: **Move 15, White: Kh1.** (a) Our position; Melanie plays Kh1 (**g1h1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#442` (ply 22) [structural distance  $d=67$ , QAP obj=52]: the master played 0-0; under  $\mu$  our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Qd1* as their *Ra1*, our *Rf1* as their *Qd1*; transferred move **g1h1** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#196` (ply 19) [structural distance  $d=51$ , QAP obj=52]: the master played 0-0; under  $\mu$  our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Qd1* as their *Ra8*, our *Rf1* as their *Qd8*; transferred move **g1h1** (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#197` (ply 19) [structural distance  $d=51$ , QAP obj=52]: the master played 0-0; under  $\mu$  our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Qd1* as their *Ra8*, our *Rf1* as their *Qd8*; transferred move **g1h1** (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kh1 (summed QAP objective 155, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



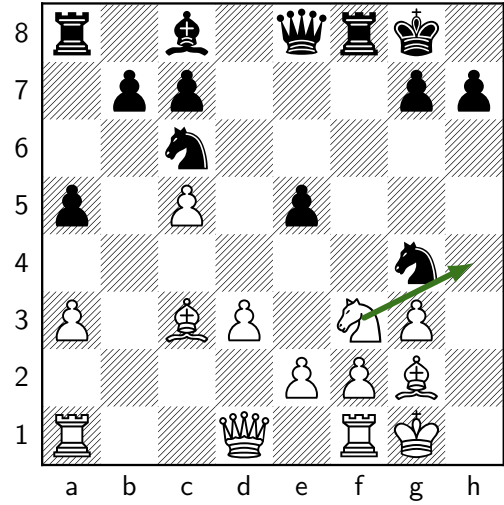
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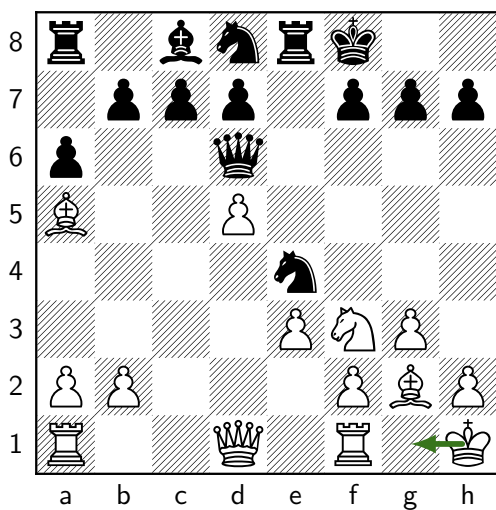


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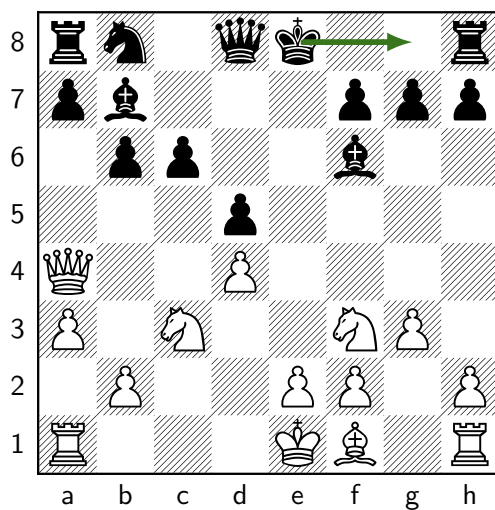


(d)

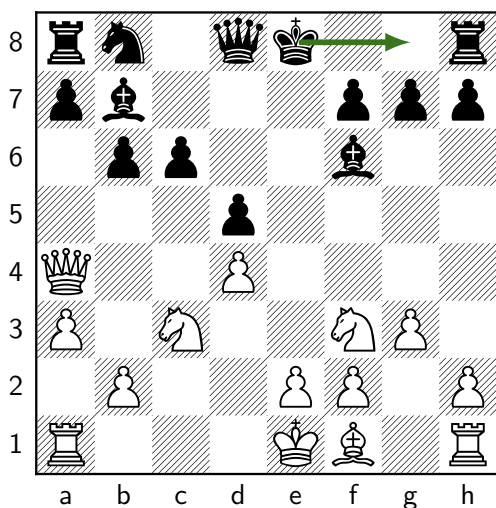
Figure 10: **Move 16, White: Ba5.** (a) Our position; Melanie plays Ba5 (c3a5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#3938` (ply 28) [structural distance  $d=81$ , QAP obj=58]: the master played Bb4; under  $\mu$  our Bc3 plays the role of that game’s Bc3; cross-type roles: our Nf3 as their Re1, our bc8 as their Pb2; transferred move c3a5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#99` (ply 23) [structural distance  $d=68$ , QAP obj=41]: the master played Be7; under  $\mu$  our Bc3 plays the role of that game’s Bf3; cross-type roles: our Qd1 as their Nb8, our bc8 as their Qd8; transferred move c3a5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#563` (ply 28) [structural distance  $d=85$ , QAP obj=35]: the master played Nh4; under  $\mu$  our Bc3 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our Pa2 as their Ra1, our Bg2 as their Rf1; transferred move c3a5 (**endorses this move**). Synthesis: 11 retrieved analogs transferred to Ba5 (summed QAP objective 295, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



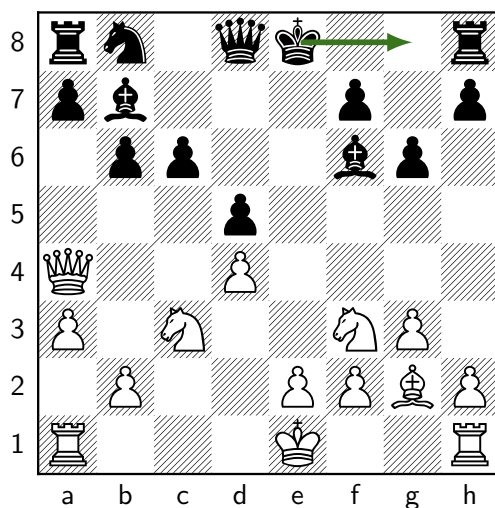
(a)



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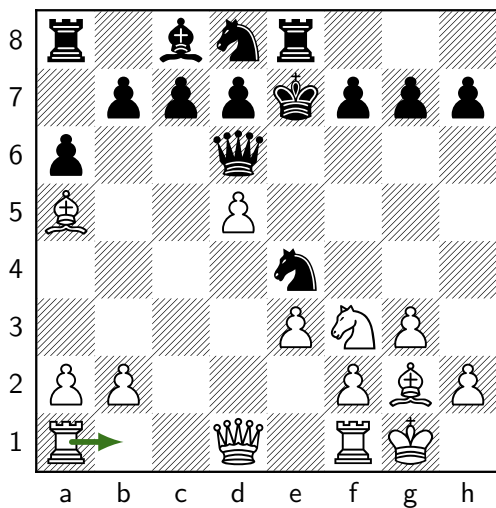


(c)

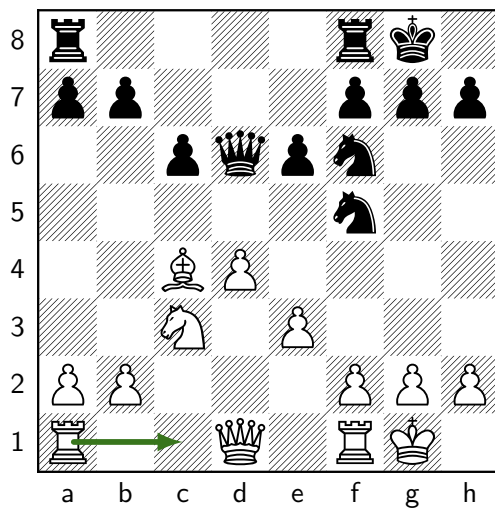


(d)

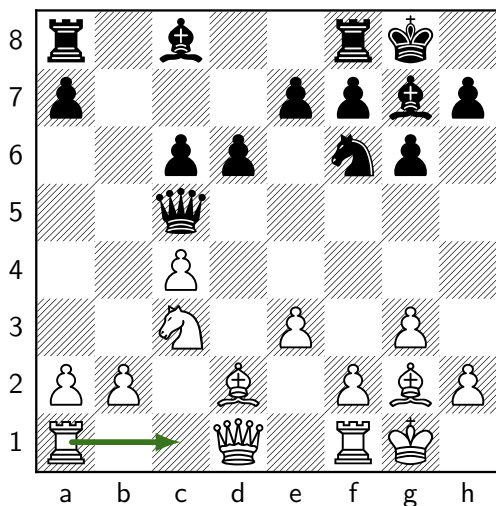
Figure 11: **Move 17, White: Kg1**. (a) Our position; Melanie plays Kg1 (**h1g1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1987` (ply 19) [structural distance  $d=93$ , QAP obj=80]: the master played 0-0; under  $\mu$  our *Kh1* plays the role of that game’s *Ke1*; cross-type roles: our *Qd1* as their *Ra8*, our *Pa2* as their *Nb8*; transferred move **h1g1 (endorses this move)**. (c) Retrieved from `ccr1_top_p0001#1955` (ply 19) [structural distance  $d=93$ , QAP obj=80]: the master played 0-0; under  $\mu$  our *Kh1* plays the role of that game’s *Ke1*; cross-type roles: our *Qd1* as their *Ra8*, our *Pa2* as their *Nb8*; transferred move **h1g1 (endorses this move)**. (d) Retrieved from `ccr1_top_p0001#1956` (ply 21) [structural distance  $d=79$ , QAP obj=79]: the master played 0-0; under  $\mu$  our *Kh1* plays the role of that game’s *Ke1*; cross-type roles: our *Qd1* as their *Ra8*, our *pg7* as their *Nb8*; transferred move **h1g1 (endorses this move)**. Synthesis: 11 retrieved analogs transferred to Kg1 (summed QAP objective 695, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



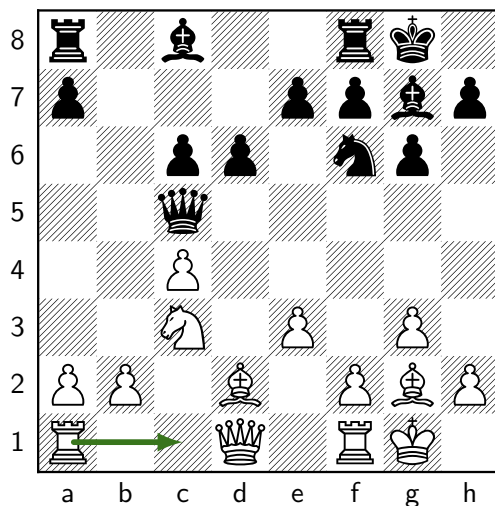
(a)



(b)

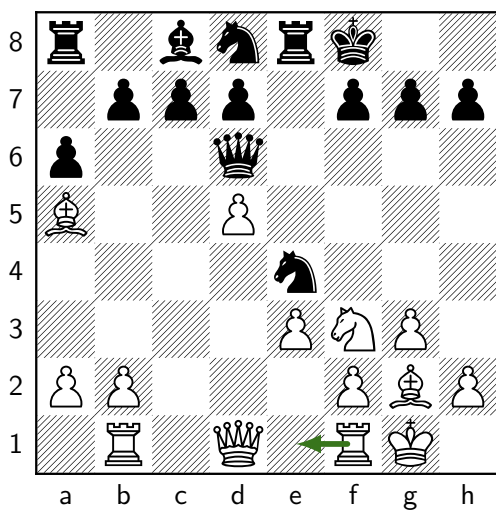


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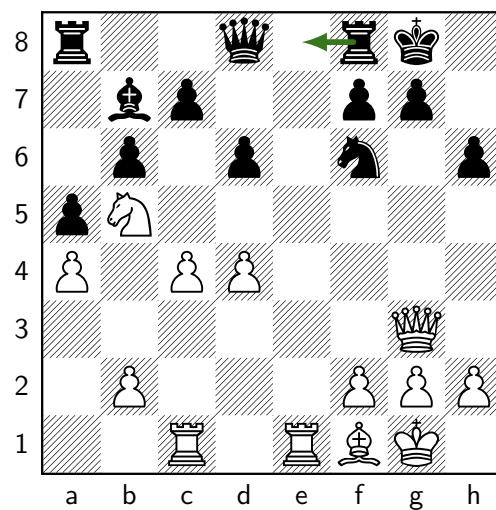


(d)

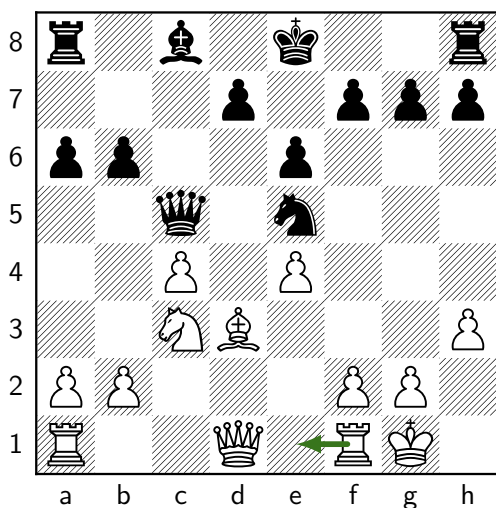
Figure 12: **Move 18, White: Rb1.** (a) Our position; Melanie plays Rb1 (a1b1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1.top_p0000#3608` (ply 24) [structural distance  $d=98$ , QAP obj=84]: the master played Rc1; under  $\mu$  our *Ra1* plays the role of that game’s *Ra1*; cross-type roles: our *Nf3* as their *Pf2*, our *bc8* as their *Nc3*; transferred move a1b1 (**endorses this move**). (c) Retrieved from `ccr1.top_p0001#3222` (ply 22) [structural distance  $d=99$ , QAP obj=82]: the master played Rc1; under  $\mu$  our *Ra1* plays the role of that game’s *Ra1*; cross-type roles: our *Nf3* as their *Rf1*, our *Bg2* as their *Pb2*; transferred move a1b1 (**endorses this move**). (d) Retrieved from `ccr1.top_p0003#2389` (ply 22) [structural distance  $d=99$ , QAP obj=82]: the master played Rc1; under  $\mu$  our *Ra1* plays the role of that game’s *Ra1*; cross-type roles: our *Nf3* as their *Rf1*, our *Bg2* as their *Pb2*; transferred move a1b1 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rb1 (summed QAP objective 248, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



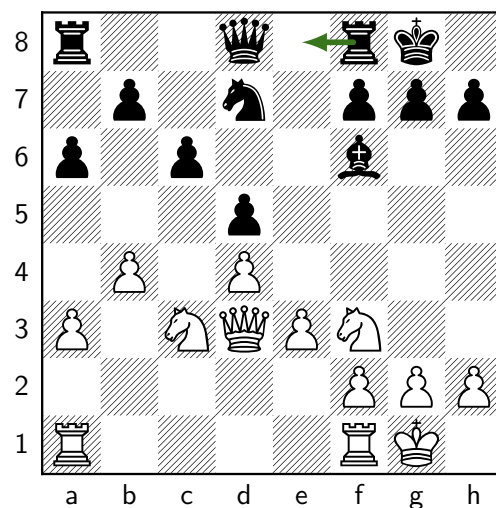
(a)



(b)

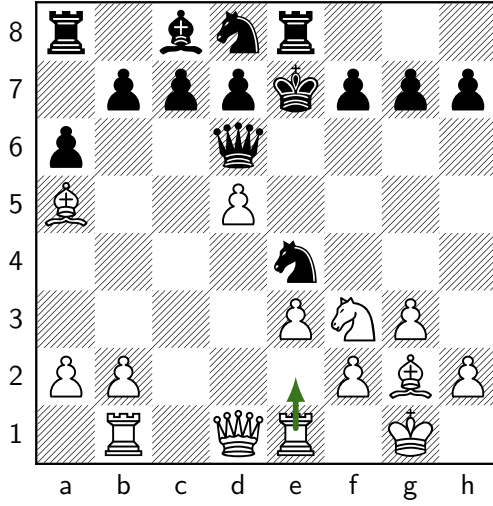


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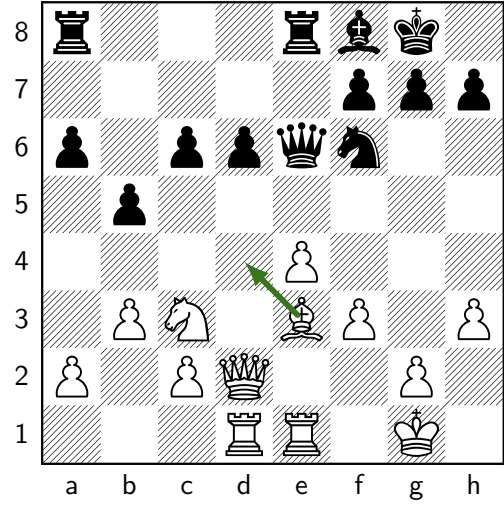


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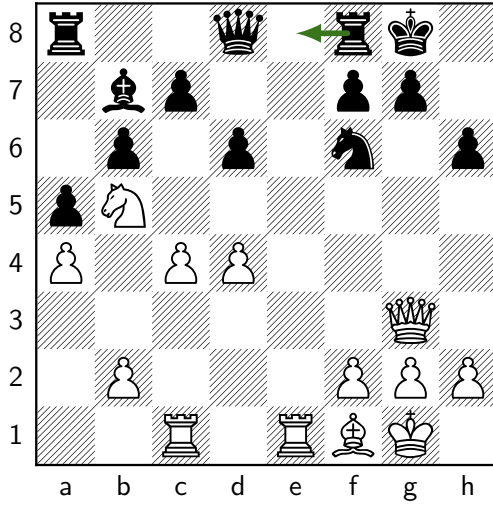
Figure 13: **Move 19, White: Re1**. (a) Our position; Melanie plays Re1 (f1e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3735` (ply 41) [structural distance  $d=90$ , QAP obj=73]: the master played Re8; under  $\mu$  our Rf1 plays the role of that game’s Rf1; cross-type roles: our Ph2 as their Bb7, our Nf3 as their Pb6; transferred move f1e1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3445` (ply 26) [structural distance  $d=90$ , QAP obj=65]: the master played Re1; under  $\mu$  our Rf1 plays the role of that game’s Rf1; cross-type roles: our Ba5 as their Nc3, our Pd5 as their Bd3; transferred move f1e1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2218` (ply 25) [structural distance  $d=85$ , QAP obj=51]: the master played Re8; under  $\mu$  our Rf1 plays the role of that game’s Rf1; cross-type roles: our Ph2 as their Ra8, our pf7 as their Nd7; transferred move f1e1 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Re1 (summed QAP objective 317, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



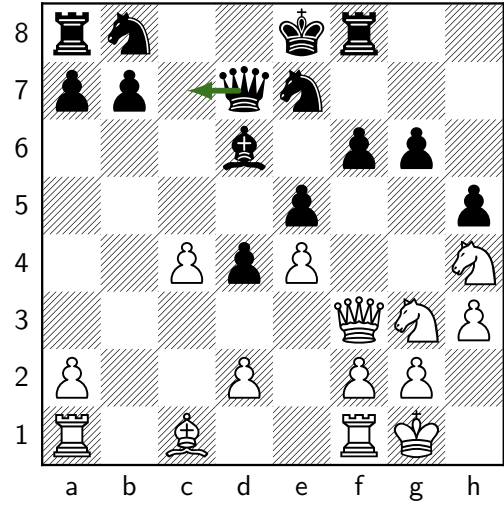
(a)



(b)



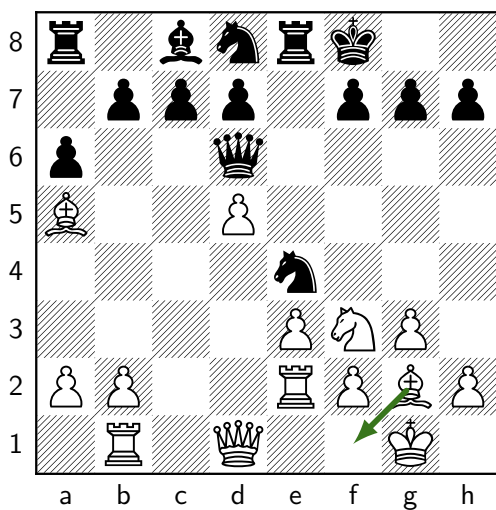
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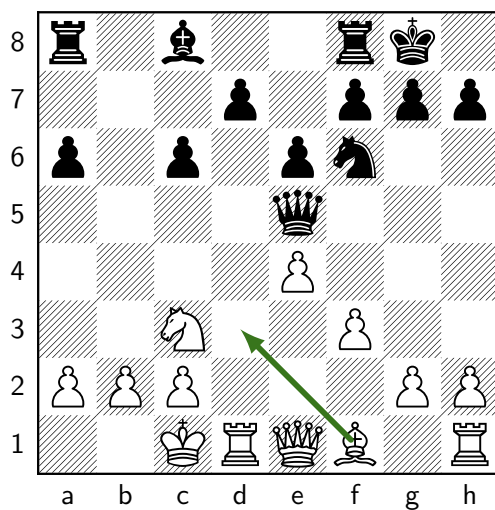
(d)

Figure 14: **Move 20, White: Re2.** (a) Our position; Melanie plays Re2 (e1e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3942` (ply 32) [structural distance  $d=104$ , QAP obj=86]: the master played Bd4; under  $\mu$  our *Re1* plays the role of that game’s *Be3* (a **cross-type** correspondence); cross-type roles: our *Qd1* as their *Rd1*, our *Nf3* as their *Re1*; transferred move e1e2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3735` (ply 41) [structural distance  $d=101$ , QAP obj=71]: the master played Re8; under  $\mu$  our *Re1* plays the role of that game’s *Rf1*; cross-type roles: our *Pe3* as their *Ra8*, our *Pa2* as their *Bb7*; transferred move e1e2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1332` (ply 29) [structural distance  $d=108$ , QAP obj=66]: the master played Qc7; under  $\mu$  our *Re1* plays the role of that game’s *Qd2* (a **cross-type** correspondence); cross-type roles: our *pg7* as their *Ra8*, our *Bg2* as their *Nb8*; transferred move e1e2 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Re2 (summed QAP objective 495, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

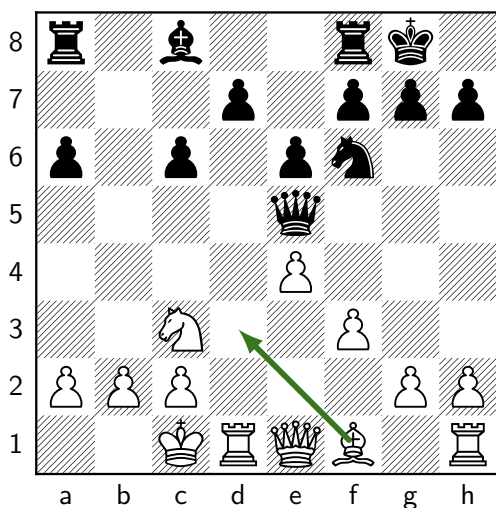




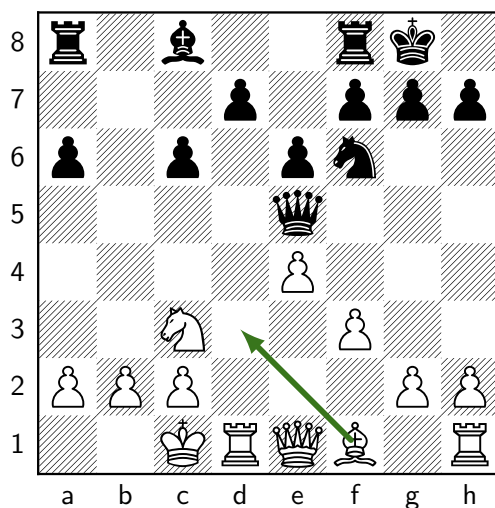
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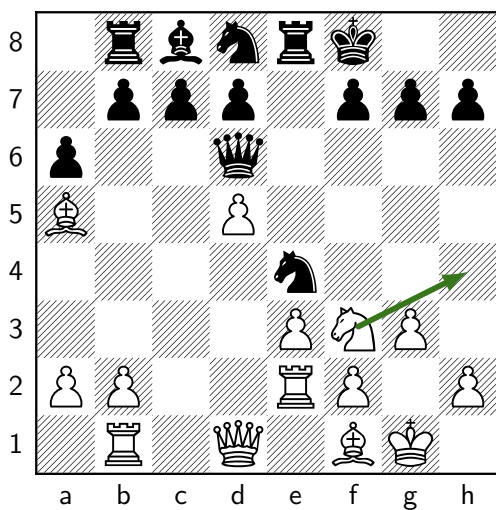
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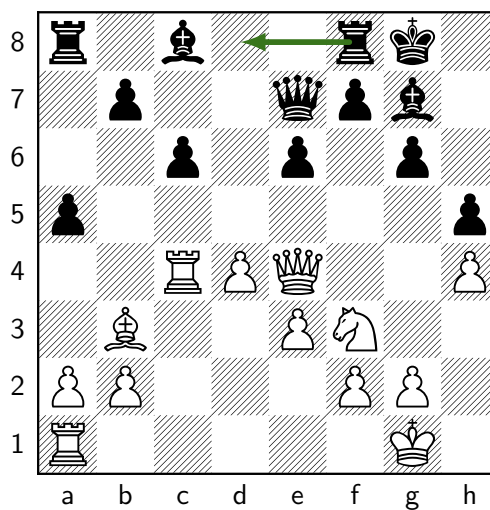
(d)

Figure 15: **Move 21, White: Bf1.** (a) Our position; Melanie plays Bf1 (g2f1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccl1_top_p0001#2078` (ply 32) [structural distance  $d=95$ , QAP obj=71]: the master played Bd3; under  $\mu$  our Bg2 plays the role of that game’s Bf1; cross-type roles: our Nf3 as their Rd1, our Rb1 as their Pa2; transferred move g2f1 (**endorses this move**). (c) Retrieved from `ccl1_top_p0002#716` (ply 32) [structural distance  $d=95$ , QAP obj=71]: the master played Bd3; under  $\mu$  our Bg2 plays the role of that game’s Bf1; cross-type roles: our Nf3 as their Rd1, our Rb1 as their Pa2; transferred move g2f1 (**endorses this move**). (d) Retrieved from `ccl1_top_p0001#2077` (ply 28) [structural distance  $d=95$ , QAP obj=71]: the master played Bd3; under  $\mu$  our Bg2 plays the role of that game’s Bf1; cross-type roles: our Nf3 as their Rd1, our Rb1 as their Pa2; transferred move g2f1 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Bf1 (summed QAP objective 327, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

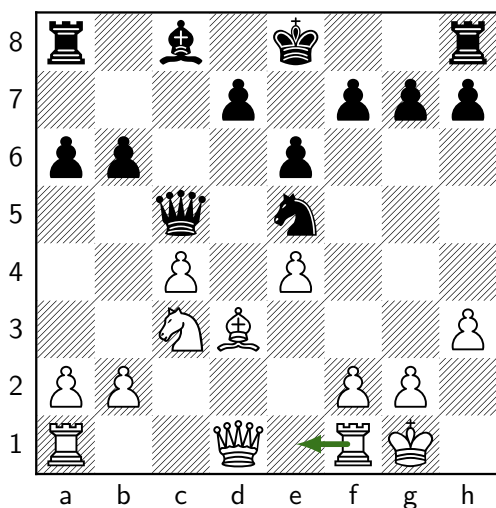




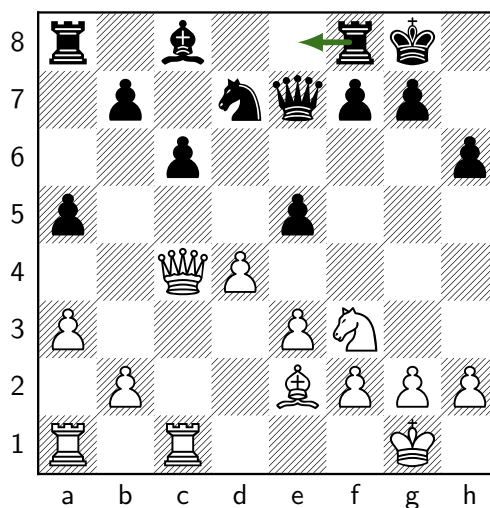
(a)



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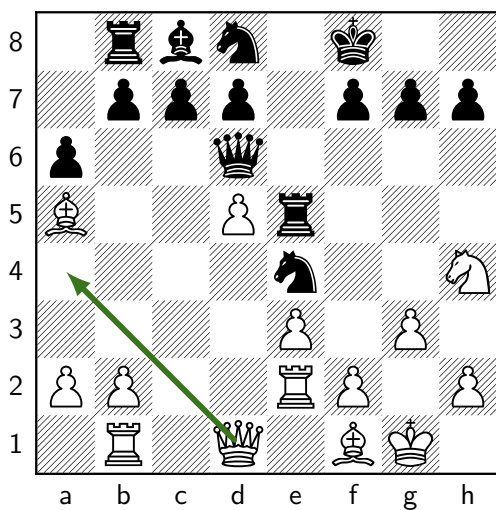


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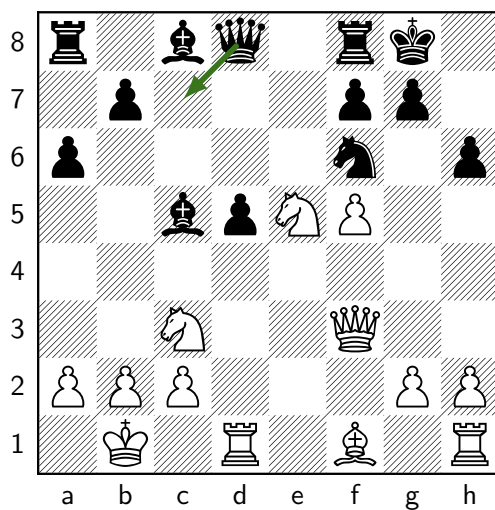


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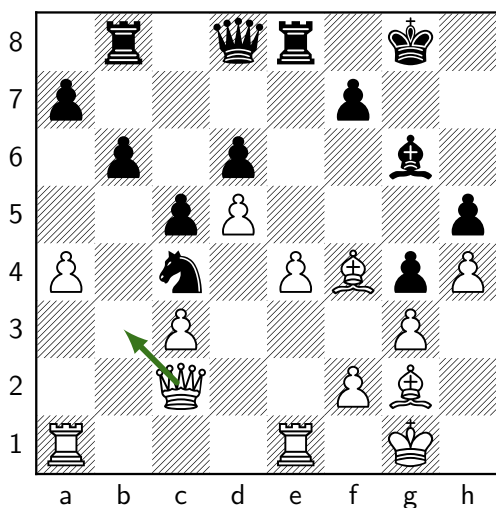
Figure 16: **Move 22, White: Nh4**. (a) Our position; Melanie plays Nh4 (f3h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2755` (ply 33) [structural distance  $d=93$ , QAP obj=90]: the master played Rd8; under  $\mu$  our Nf3 plays the role of that game’s Rf1 (a **cross-type** correspondence); cross-type roles: our Ph2 as their Bc8, our nd8 as their Pb7; transferred move f3h4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3445` (ply 26) [structural distance  $d=94$ , QAP obj=63]: the master played Re1; under  $\mu$  our Nf3 plays the role of that game’s Rf1 (a **cross-type** correspondence); cross-type roles: our pb7 as their Ra1, our rb8 as their Nc3; transferred move f3h4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3307` (ply 29) [structural distance  $d=85$ , QAP obj=63]: the master played Re8; under  $\mu$  our Nf3 plays the role of that game’s Rf1 (a **cross-type** correspondence); cross-type roles: our rb8 as their Bc8, our Qd1 as their Nd7; transferred move f3h4 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Nh4 (summed QAP objective 295, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



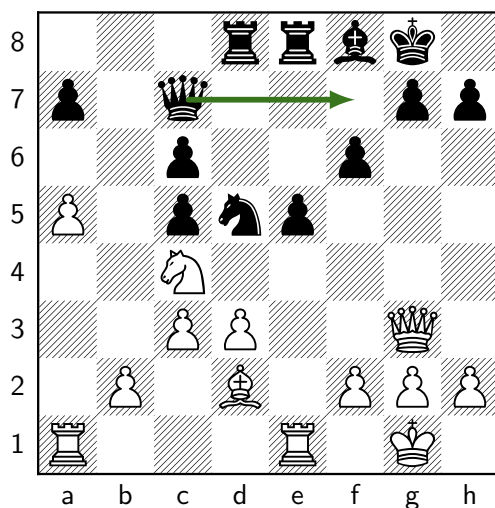
(a)



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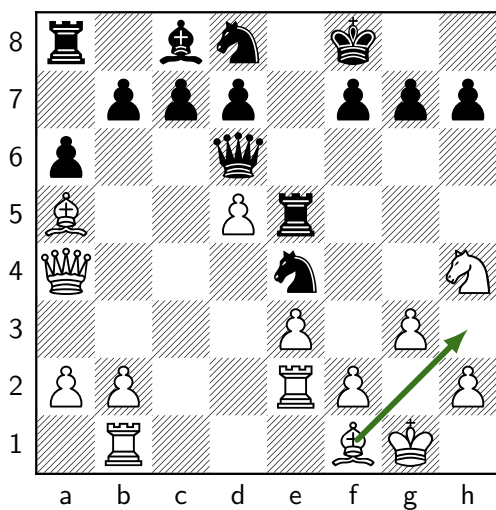


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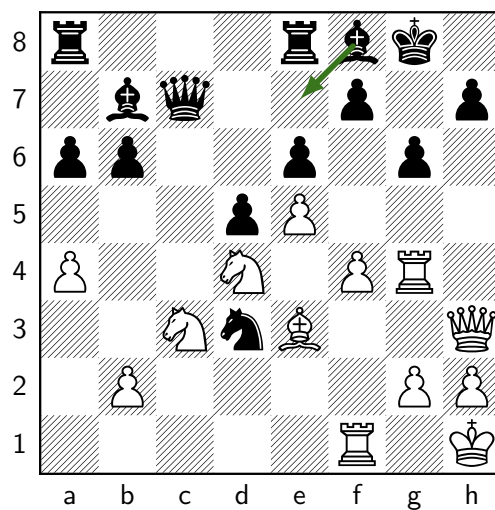


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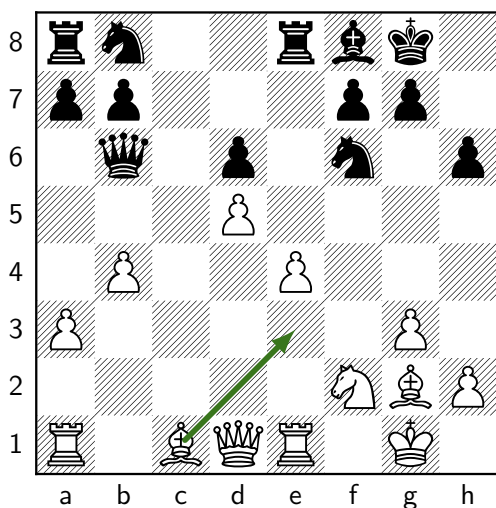
Figure 17: **Move 23, White: Qa4.** (a) Our position; Melanie plays Qa4 (d1a4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#2139` (ply 29) [structural distance  $d=86$ , QAP obj=80]: the master played Qc7; under  $\mu$  our Qd1 plays the role of that game’s Qd1; cross-type roles: our nd8 as their Pb7, our Nh4 as their Pa6; transferred move d1a4 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#2886` (ply 44) [structural distance  $d=90$ , QAP obj=77]: the master played Qb3; under  $\mu$  our Qd1 plays the role of that game’s Qc2; cross-type roles: our Rb1 as their Bg2, our Nh4 as their Pc3; transferred move d1a4 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#2072` (ply 37) [structural distance  $d=89$ , QAP obj=105]: the master played Qf7; under  $\mu$  our Ba5 plays the role of that game’s Qc2 (a **cross-type** correspondence); cross-type roles: our Qd1 as their Re8, our Nh4 as their Ph7; transferred move a5e1. Synthesis: 2 retrieved analogs transferred to Qa4 (summed QAP objective 157, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



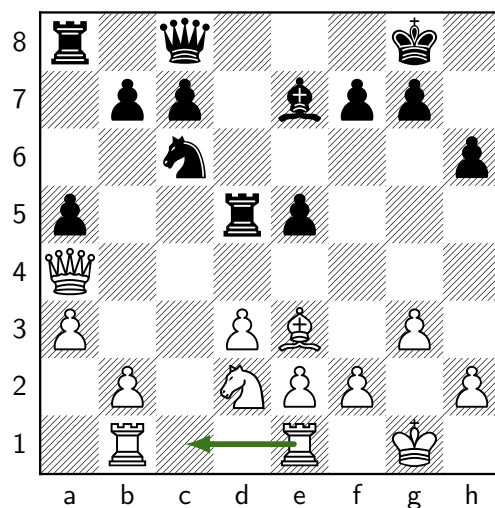
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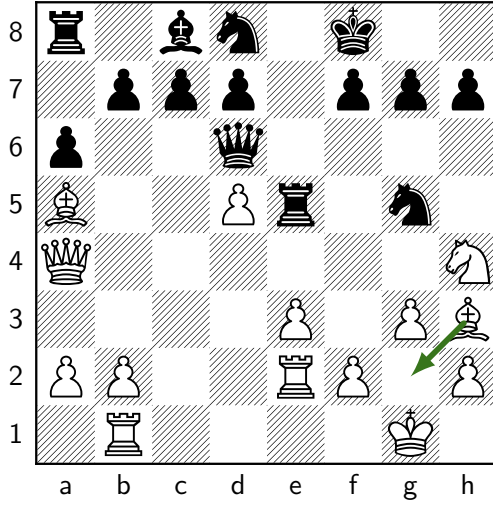


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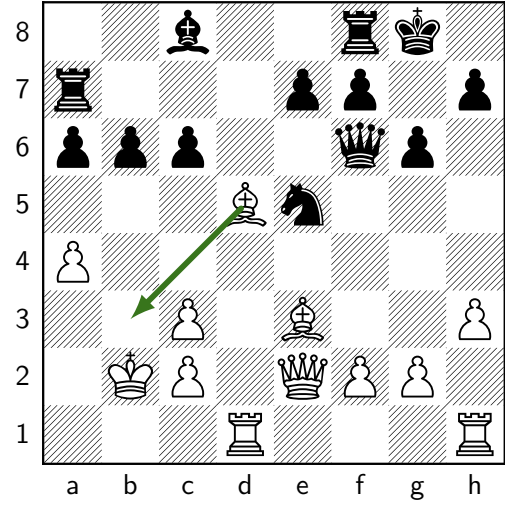


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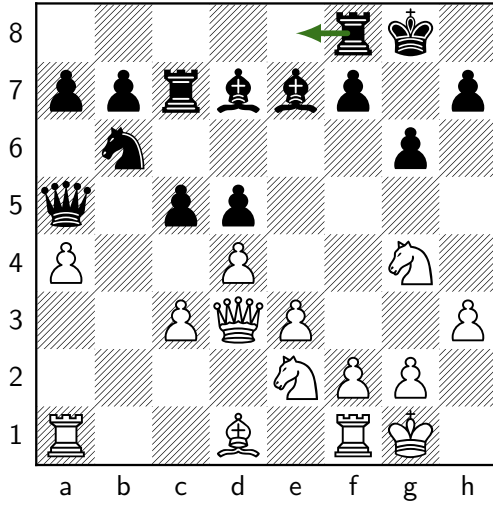
Figure 18: **Move 24, White: Bh3.** (a) Our position; Melanie plays Bh3 (f1h3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1544` (ply 45) [structural distance  $d=87$ , QAP obj=91]: the master played Be7; under  $\mu$  our Bf1 plays the role of that game’s Bf1; cross-type roles: our Pa2 as their Bb7, our Nh4 as their Ph7; transferred move f1h3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#138` (ply 34) [structural distance  $d=93$ , QAP obj=66]: the master played Be3; under  $\mu$  our Bf1 plays the role of that game’s Bc1; cross-type roles: our Nh4 as their Ra1, our Rb1 as their Qd1; transferred move f1h3 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#1397` (ply 32) [structural distance  $d=92$ , QAP obj=64]: the master played Rec1; under  $\mu$  our Bf1 plays the role of that game’s Re1 (a **cross-type** correspondence); cross-type roles: our ne4 as their Pb2, our qd6 as their Be3; transferred move f1h3 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Bh3 (summed QAP objective 222, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



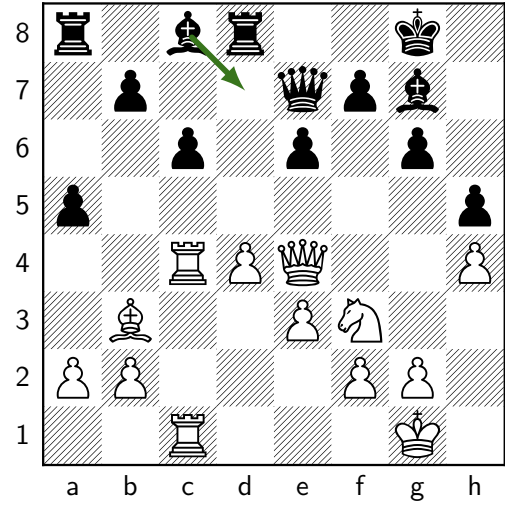
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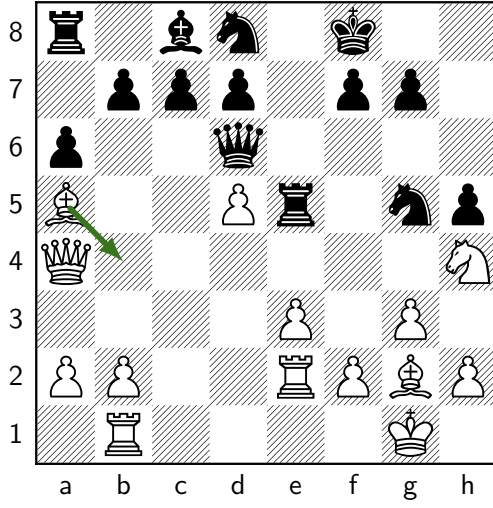


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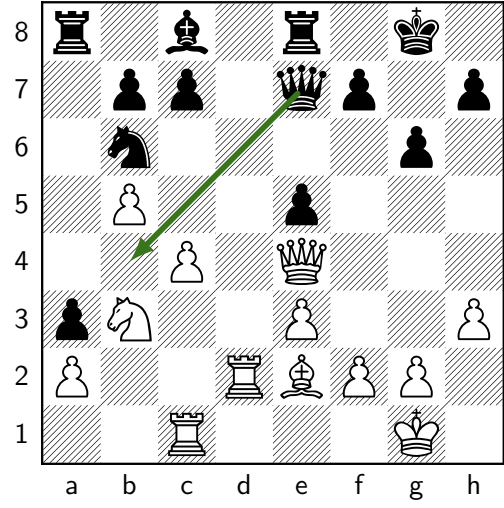


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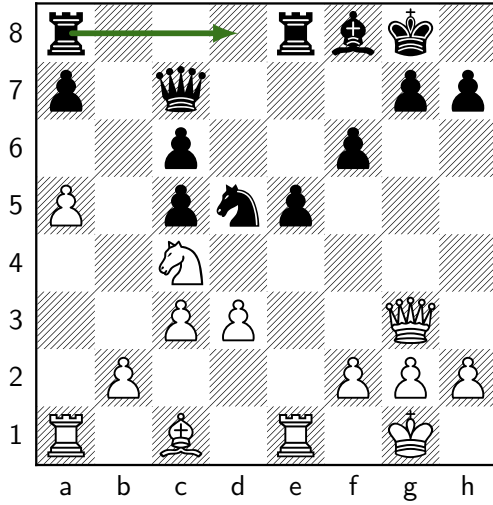
Figure 19: **Move 25, White: Bg2.** (a) Our position; Melanie plays Bg2 (h3g2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3447` (ply 36) [structural distance  $d=124$ , QAP obj=47]: the master played Bb3; under  $\mu$  our Bh3 plays the role of that game’s Bd5; cross-type roles: our Nh4 as their Rd1, our Re2 as their Pg2; transferred move h3g2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1672` (ply 41) [structural distance  $d=123$ , QAP obj=114]: the master played Re8; under  $\mu$  our Pf2 plays the role of that game’s Rf1 (a **cross-type** correspondence); cross-type roles: our Ba5 as their Rc7, our qd6 as their Bd7; transferred move f2f4. (d) Retrieved from `ccrl_top_p0002#2755` (ply 35) [structural distance  $d=116$ , QAP obj=80]: the master played Bd7; under  $\mu$  our Pf2 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our Pa2 as their Ra8, our Bh3 as their Rd8; transferred move f2f4. Synthesis: 1 retrieved analog transferred to Bg2 (summed QAP objective 47, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



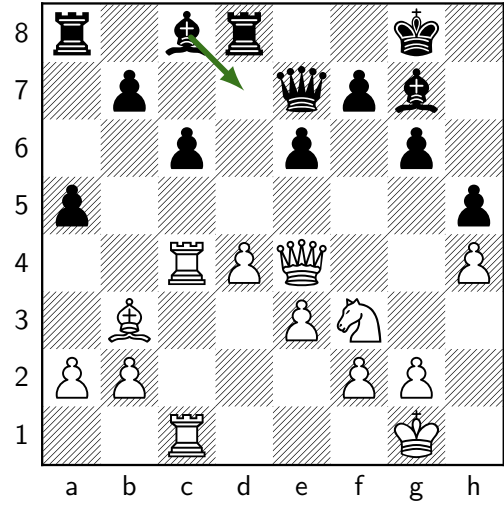
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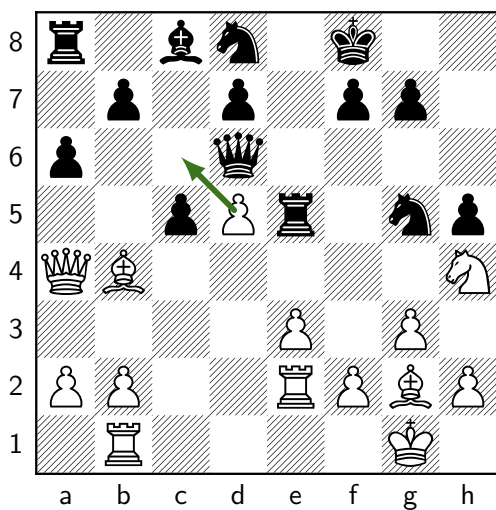


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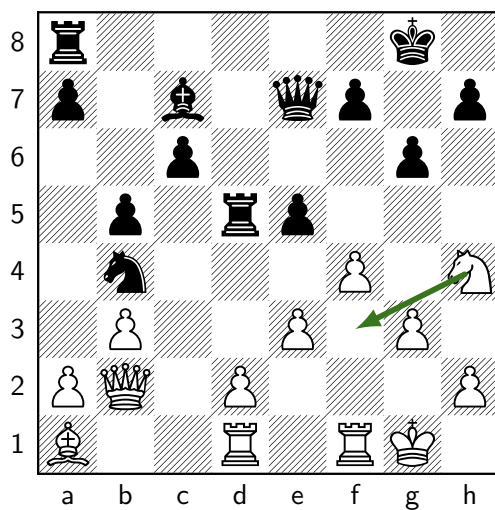


(d)

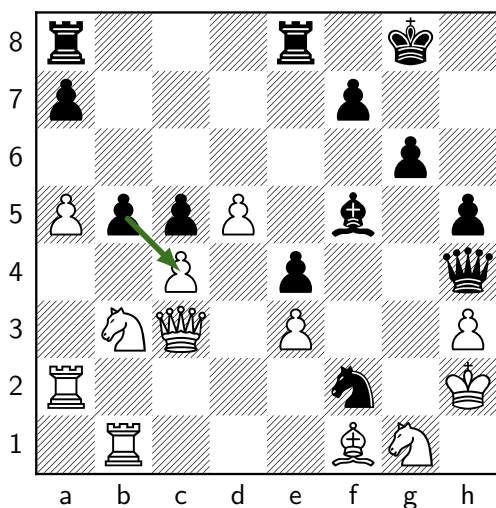
Figure 20: **Move 26, White: Bb4.** (a) Our position; Melanie plays Bb4 (a5b4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3977` (ply 45) [structural distance  $d=112$ , QAP obj=126]: the master played Qb4; under  $\mu$  our *Re2* plays the role of that game’s *Qe2* (a **cross-type** correspondence); cross-type roles: our *ph5* as their *Ra8*, our *Qa4* as their *Pc7*; transferred move e2c2. (c) Retrieved from `ccr1_top_p0001#2072` (ply 35) [structural distance  $d=107$ , QAP obj=122]: the master played Rad8; under  $\mu$  our *Pg3* plays the role of that game’s *Ra1* (a **cross-type** correspondence); cross-type roles: our *Pa2* as their *Bf8*, our *Rb1* as their *Pa7*; transferred move g3g4. (d) Retrieved from `ccr1_top_p0002#2755` (ply 35) [structural distance  $d=98$ , QAP obj=112]: the master played Bd7; under  $\mu$  our *Bg2* plays the role of that game’s *Bc1*; cross-type roles: our *pb7* as their *Ra8*, our *Qa4* as their *Rd8*; transferred move g2f3. Synthesis: no retrieved analog transferred to Bb4 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



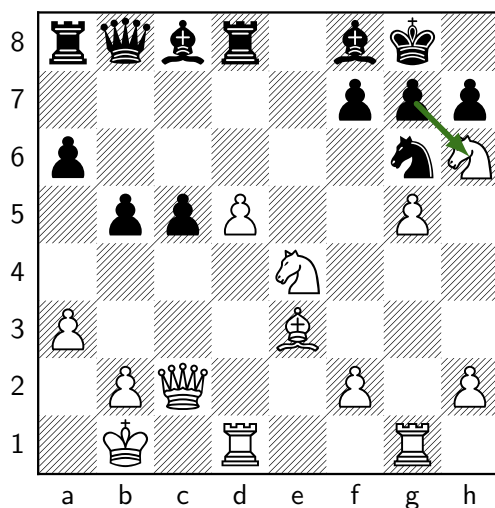
(a)



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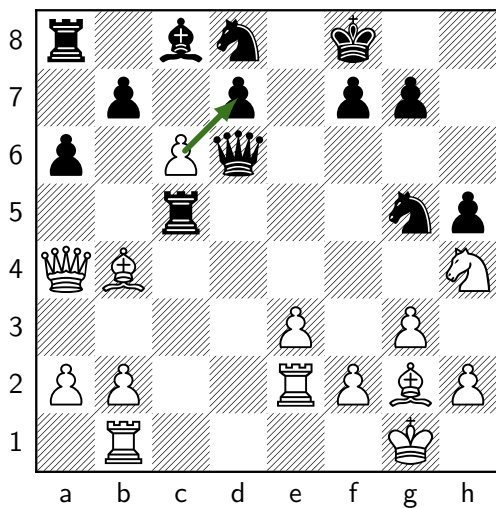


(c)

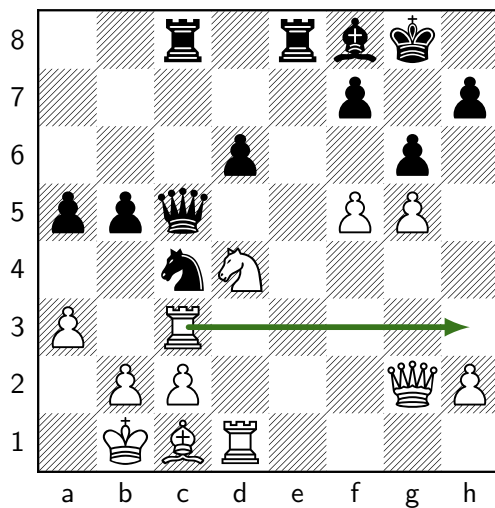


(d)

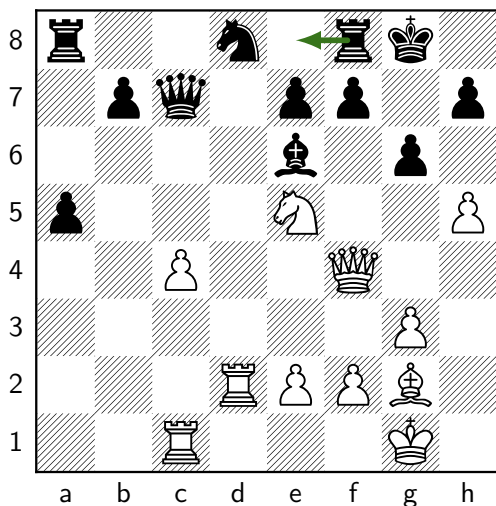
Figure 21: **Move 27, White: dxc6**. (a) Our position; Melanie plays dxc6 (d5c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1481` (ply 50) [structural distance  $d=118$ , QAP obj=40]: the master played Nf3; under  $\mu$  our Pd5 plays the role of that game’s Nh4 (a **cross-type** correspondence); cross-type roles: our Nh4 as their Ba1, our Pb2 as their Rf1; transferred move d5c6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1476` (ply 53) [structural distance  $d=112$ , QAP obj=38]: the master played bxc4; under  $\mu$  our Pd5 plays the role of that game’s Pb4; cross-type roles: our Nh4 as their Ra8, our Bg2 as their Pg6; transferred move d5c6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1895` (ply 55) [structural distance  $d=120$ , QAP obj=25]: the master played gxh6; under  $\mu$  our Pd5 plays the role of that game’s Pg2; cross-type roles: our ra8 as their Qb8, our Qa4 as their Rd8; transferred move d5c6 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to dxc6 (summed QAP objective 126, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



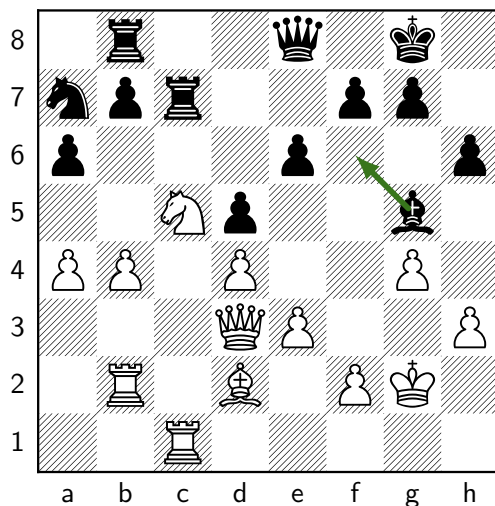
(a)



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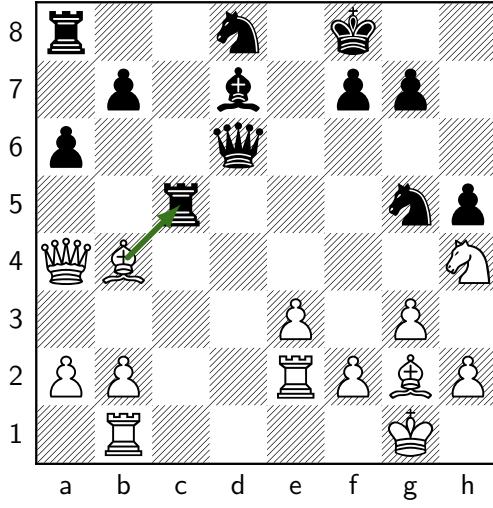
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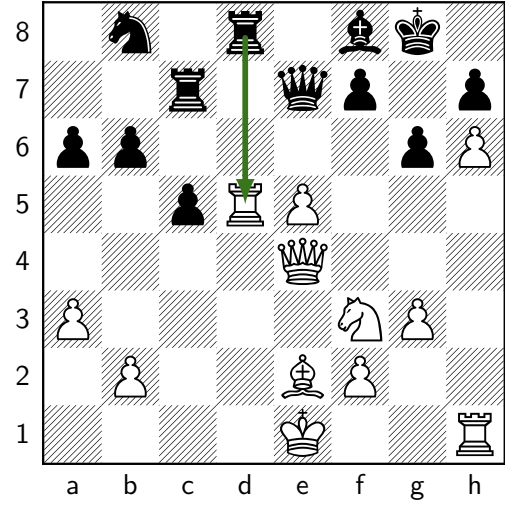
(d)

Figure 22: **Move 28, White: cxd7**. (a) Our position; Melanie plays cxd7 (c6d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1248` (ply 52) [structural distance  $d=132$ , QAP obj=99]: the master played  $Rh3$ ; under  $\mu$  our  $Bb4$  plays the role of that game's  $Rc3$  (a **cross-type** correspondence); cross-type roles: our  $Qa4$  as their  $Bc1$ , our  $pa6$  as their  $Qg2$ ; transferred move  $b4a5$ . (c) Retrieved from `ccr1_top_p0003#2795` (ply 43) [structural distance  $d=129$ , QAP obj=66]: the master played  $Re8$ ; under  $\mu$  our  $Ph2$  plays the role of that game's  $Rf1$  (a **cross-type** correspondence); cross-type roles: our  $Bg2$  as their  $Nd8$ , our  $Nh4$  as their  $Pb7$ ; transferred move  $h2h3$ . (d) Retrieved from `ccr1_top_p0001#1885` (ply 69) [structural distance  $d=129$ , QAP obj=64]: the master played  $Bf6$ ; under  $\mu$  our  $Pg3$  plays the role of that game's  $Bg4$  (a **cross-type** correspondence); cross-type roles: our  $Qa4$  as their  $Rb8$ , our  $Rb1$  as their  $Qe8$ ; transferred move  $g3g4$ . Synthesis: no retrieved analog transferred to cxd7 – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.

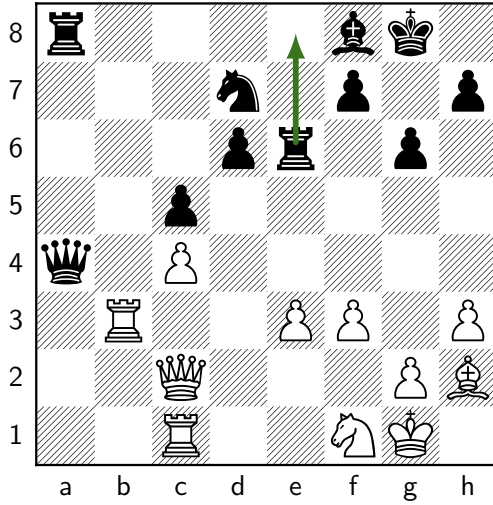




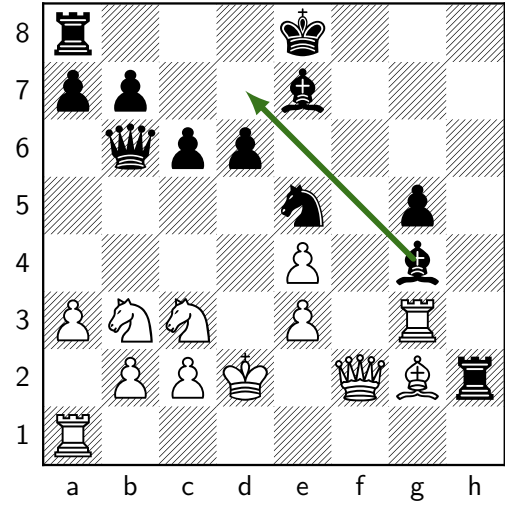
(a)



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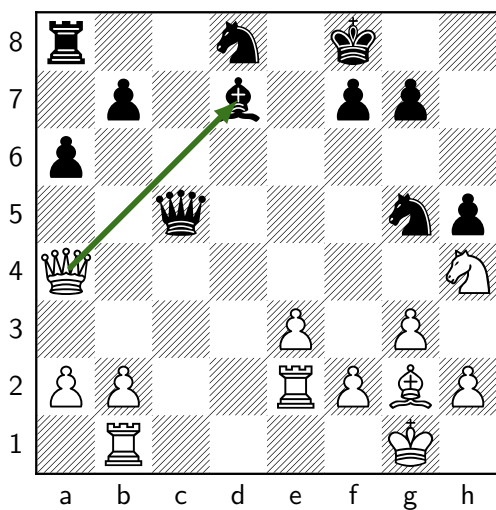


(c)

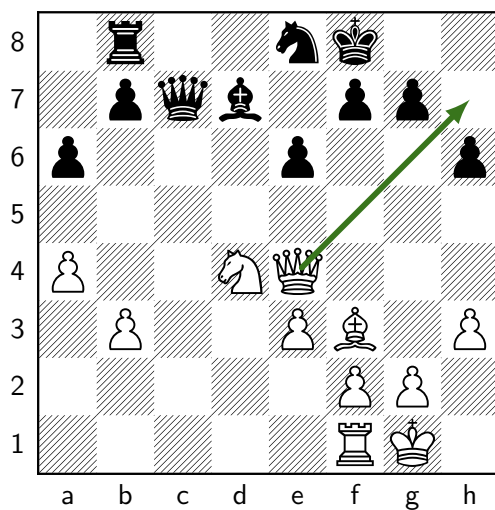


(d)

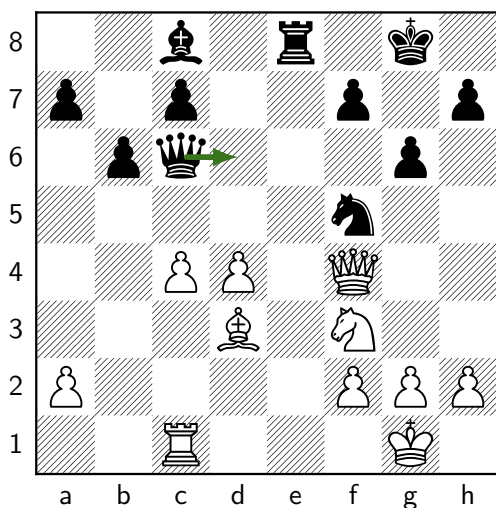
Figure 23: **Move 29, White: Bxc5.** (a) Our position; Melanie plays Bxc5 (b4c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1242` (ply 45) [structural distance  $d=116$ , QAP obj=15]: the master played Rxd5; under  $\mu$  our *Bb4* plays the role of that game’s *Rd1* (a **cross-type** correspondence); cross-type roles: our *pb7* as their *Nb8*, our *Nh4* as their *Bf8*; transferred move **b4c5 (endorses this move)**. (c) Retrieved from `ccr1_top_p0000#221` (ply 61) [structural distance  $d=111$ , QAP obj=58]: the master played Ree8; under  $\mu$  our *Bg2* plays the role of that game’s *Re3* (a **cross-type** correspondence); cross-type roles: our *Qa4* as their *Ra8*, our *Pb2* as their *Bf8*; transferred move **g2f1**. (d) Retrieved from `ccr1_top_p0003#1374` (ply 41) [structural distance  $d=107$ , QAP obj=51]: the master played Bd7; under  $\mu$  our *Qa4* plays the role of that game’s *Bg5* (a **cross-type** correspondence); cross-type roles: our *ra8* as their *Pb7*, our *Pg3* as their *Be7*; transferred move **a4a3**. Synthesis: 1 retrieved analog transferred to Bxc5 (summed QAP objective 15, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



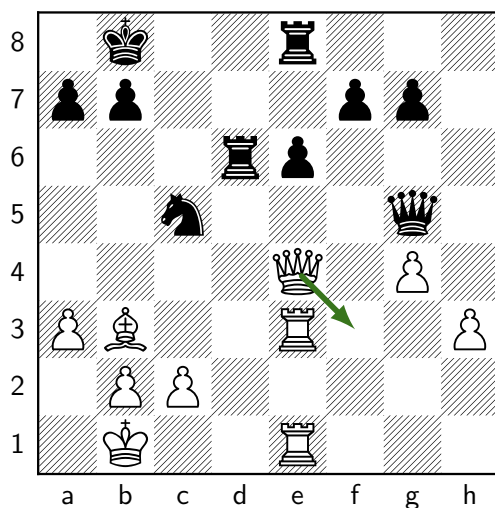
(a)



(b)

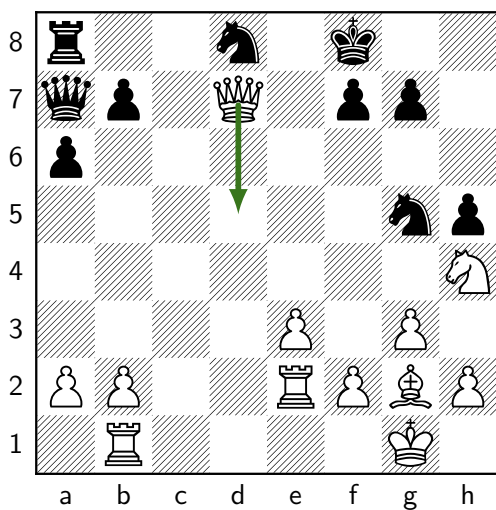


(c)

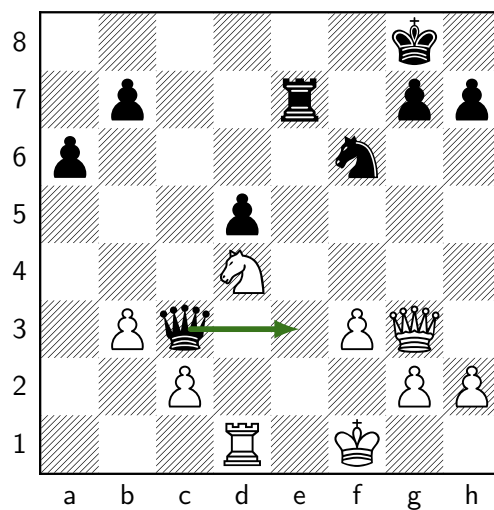


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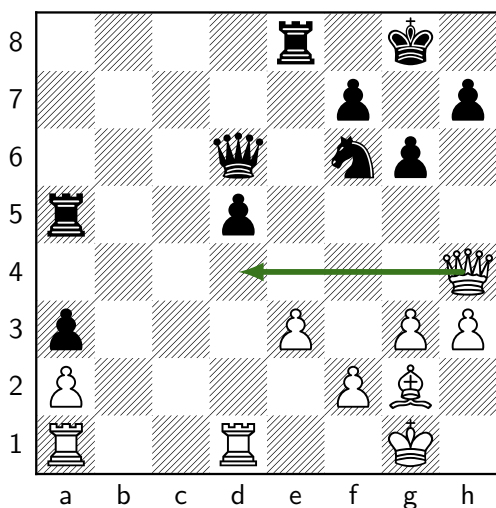
Figure 24: **Move 30, White: Qxd7.** (a) Our position; Melanie plays Qxd7 (a4d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2864` (ply 50) [structural distance  $d=99$ , QAP obj=37]: the master played Qh7; under  $\mu$  our Qa4 plays the role of that game’s Qe4; cross-type roles: our nd8 as their Pb3, our Nh4 as their Bf3; transferred move a4d7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#584` (ply 53) [structural distance  $d=102$ , QAP obj=66]: the master played Qd6; under  $\mu$  our Qa4 plays the role of that game’s Qc3; cross-type roles: our Ph2 as their Bc8, our Nh4 as their Pg6; transferred move a4e4. (d) Retrieved from `ccr1_top_p0000#3614` (ply 64) [structural distance  $d=103$ , QAP obj=62]: the master played Qf3; under  $\mu$  our Qa4 plays the role of that game’s Qe4; cross-type roles: our Qa4 as their Bb3, our Qa4 as their Re3; transferred move a4c4. Synthesis: 1 retrieved analog transferred to Qxd7 (summed QAP objective 37, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



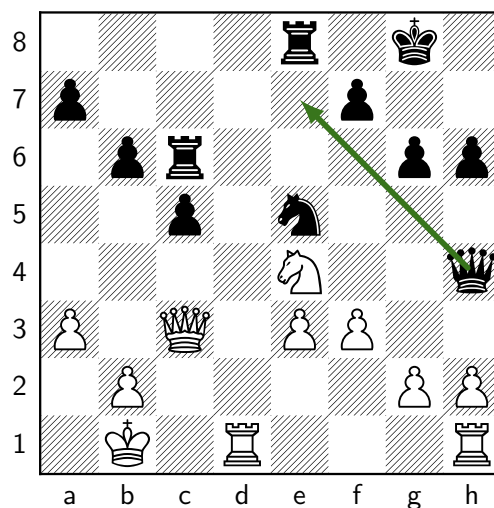
(a)



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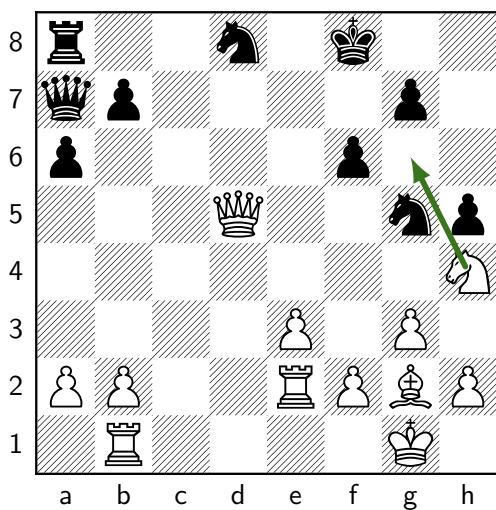


(c)

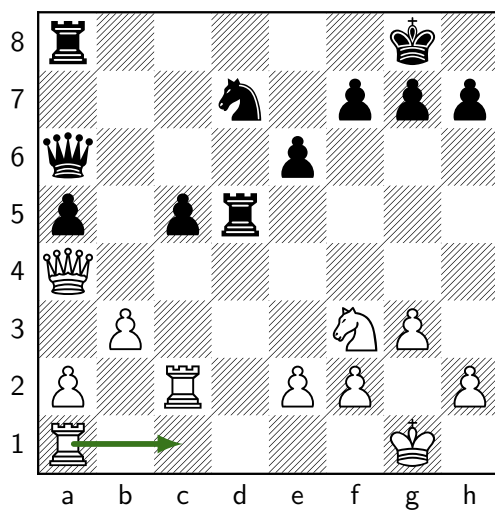


(d)

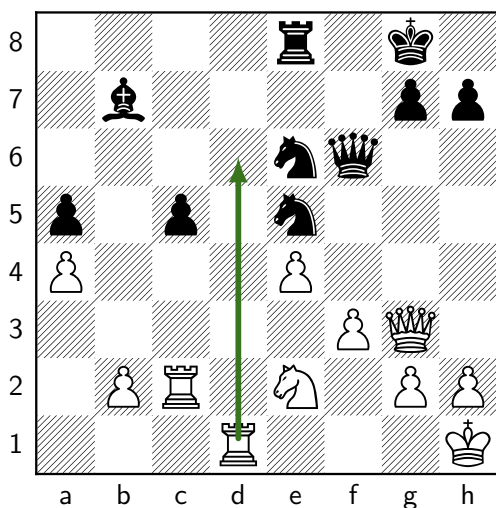
Figure 25: **Move 31, White: Qd5.** (a) Our position; Melanie plays Qd5 (d7d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#800` (ply 61) [structural distance  $d=104$ , QAP obj=87]: the master played Qe3; under  $\mu$  our Qd7 plays the role of that game's Qc6; cross-type roles: our Rb1 as their Nf6, our Bg2 as their Pd5; transferred move d7d5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2616` (ply 50) [structural distance  $d=99$ , QAP obj=34]: the master played Qd4; under  $\mu$  our Qd7 plays the role of that game's Qh4; cross-type roles: our Bg2 as their Rd1, our Rb1 as their Bg2; transferred move d7d5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2722` (ply 51) [structural distance  $d=108$ , QAP obj=90]: the master played Qe7; under  $\mu$  our Qd7 plays the role of that game's Qh5; cross-type roles: our ng5 as their Pb6, our qa7 as their Rc6; transferred move d7h3. Synthesis: 2 retrieved analogs transferred to Qd5 (summed QAP objective 121, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



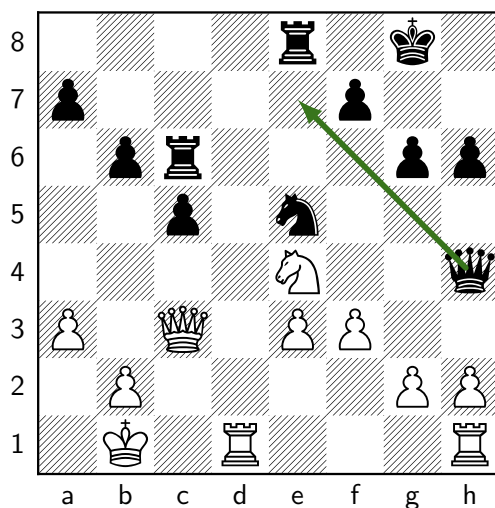
(a)



(b)



(c)



(d)

Figure 26: **Move 32, White: Ng6+.** (a) Our position; Melanie plays Ng6+ (h4g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2180` (ply 50) [structural distance  $d=100$ , QAP obj=67]: the master played Rac1; under  $\mu$  our Nh4 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Bg2 as their Pa2; transferred move h4g6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3000` (ply 62) [structural distance  $d=96$ , QAP obj=51]: the master played Rd6; under  $\mu$  our Nh4 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our Bg2 as their Ne2, our qa7 as their ne6; transferred move h4g6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2722` (ply 51) [structural distance  $d=97$ , QAP obj=107]: the master played Qe7; under  $\mu$  our Qd5 plays the role of that game’s Qh5; cross-type roles: our Nh4 as their Pb6, our Bg2 as their pa3; transferred move d5b3. Synthesis: 2 retrieved analogs transferred to Ng6+ (summed QAP objective 117, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

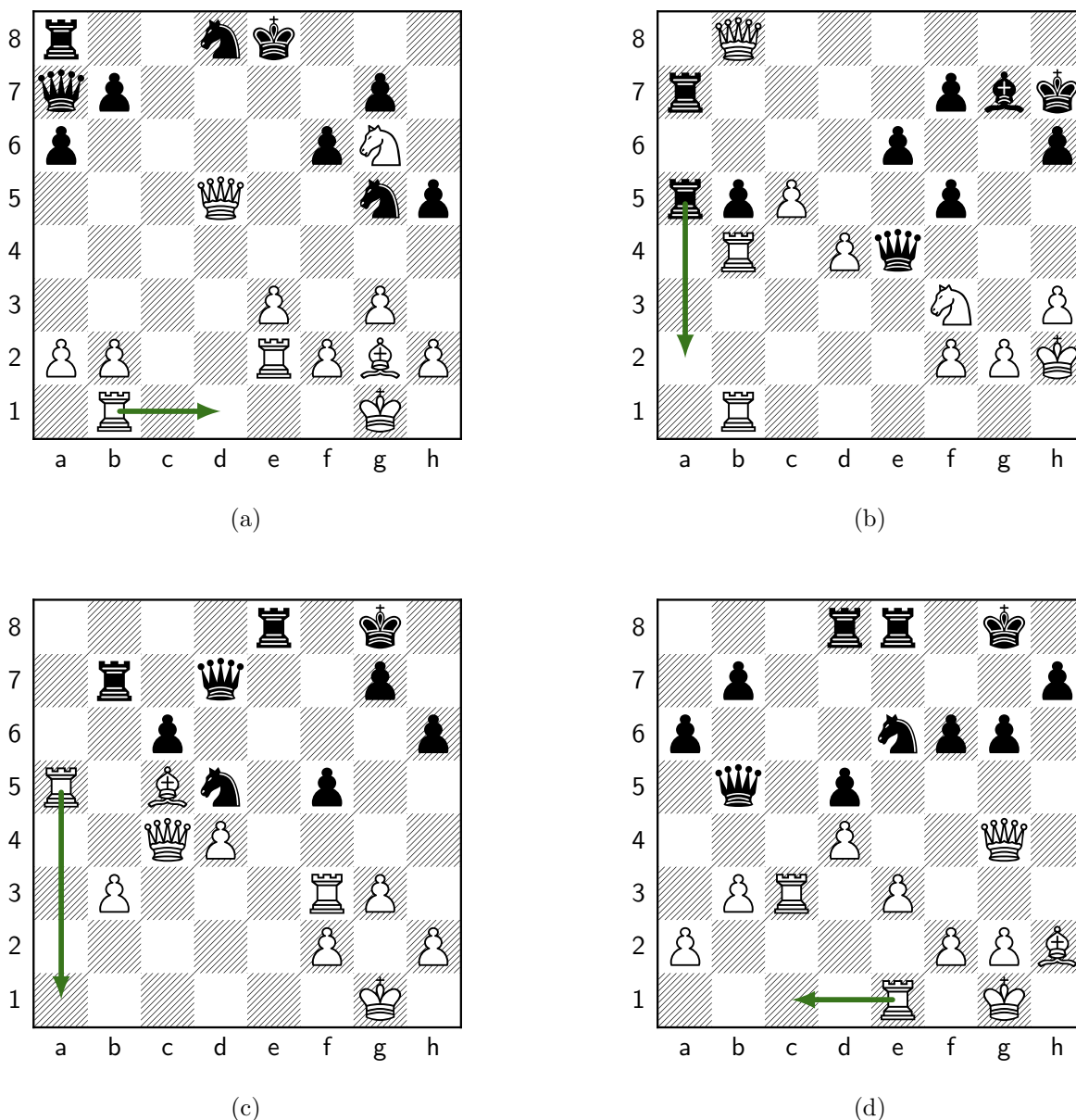
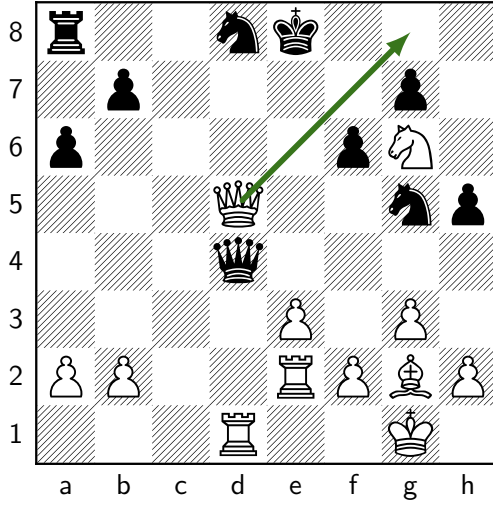
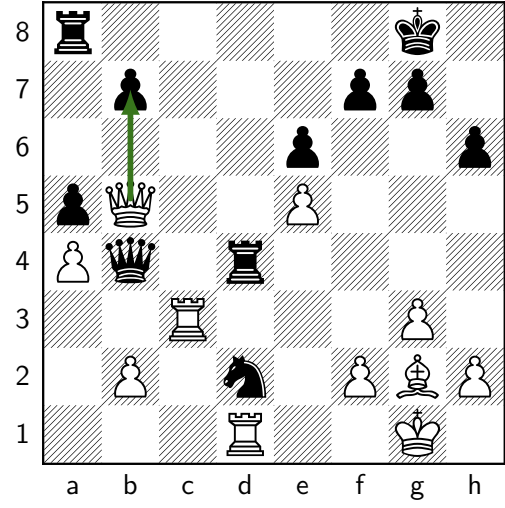


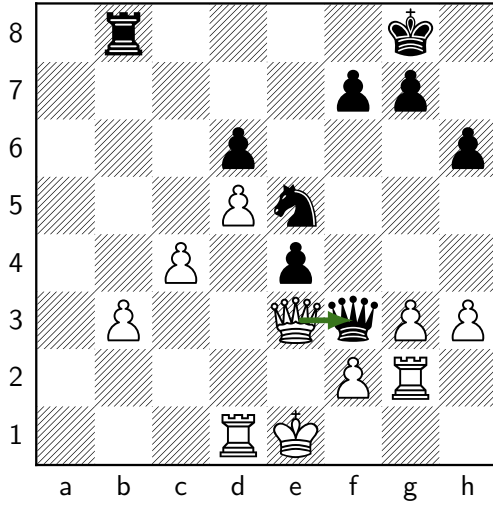
Figure 27: **Move 33, White: Rd1**. (a) Our position; Melanie plays **Rd1** (**b1d1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#2639` (ply 77) [structural distance  $d=100$ , QAP obj=45]: the master played **Ra2**; under  $\mu$  our **Rb1** plays the role of that game's **Ra4**; cross-type roles: our **Pg3** as their **qb8**, our **ph5** as their **Ra7**; transferred move **b1d1** (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#1216` (ply 54) [structural distance  $d=99$ , QAP obj=15]: the master played **Ra1**; under  $\mu$  our **Rb1** plays the role of that game's **Ra5**; cross-type roles: our **Bg2** as their **Pb3**, our **Ng6** as their **Pg3**; transferred move **b1d1** (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#596` (ply 62) [structural distance  $d=102$ , QAP obj=67]: the master played **Rec1**; under  $\mu$  our **Rb1** plays the role of that game's **Re1**; cross-type roles: our **Bg2** as their **Pg2**, our **Pg3** as their **Bh2**; transferred move **b1e1**. Synthesis: 3 retrieved analogs transferred to **Rd1** (summed QAP objective 73, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



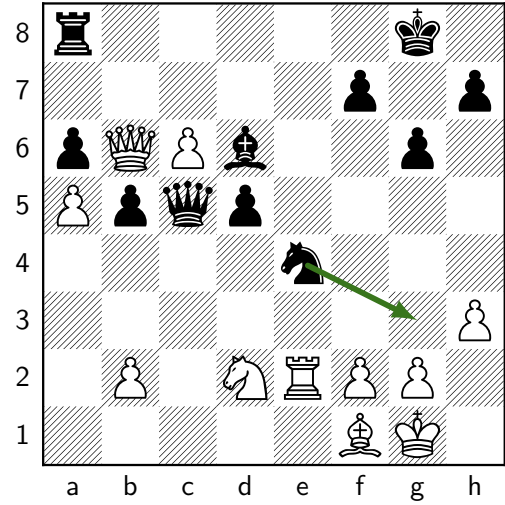
(a)



(b)

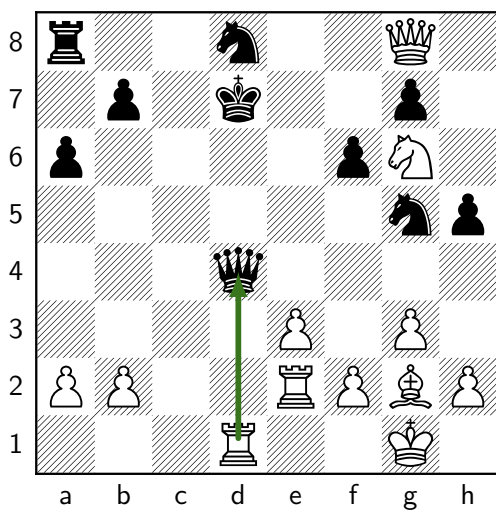


(c)

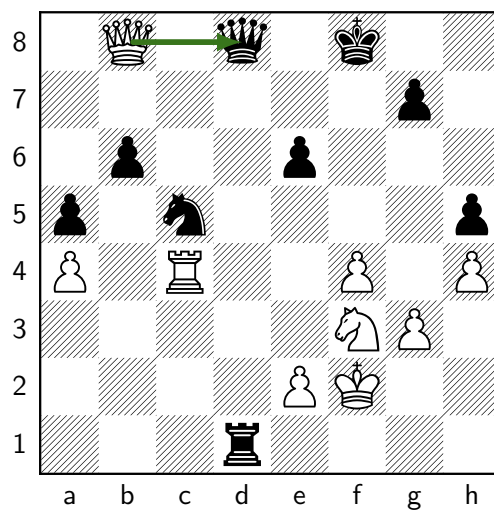


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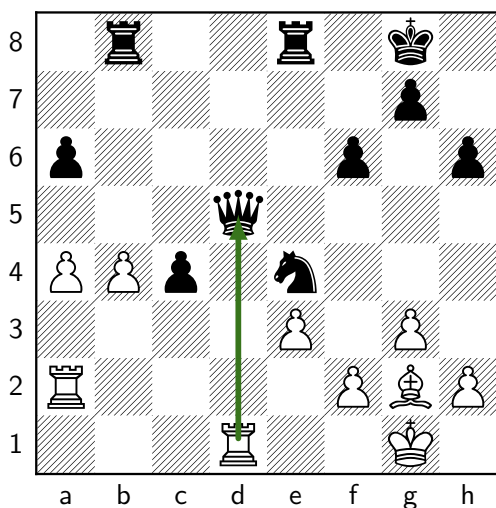
Figure 28: **Move 34, White: Qg8+.** (a) Our position; Melanie plays Qg8+ (d5g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3314` (ply 54) [structural distance  $d=109$ , QAP obj=76]: the master played Qxb7; under  $\mu$  our Qd5 plays the role of that game’s Qb5; cross-type roles: our qd4 as their nd2, our Re2 as their Bg2; transferred move d5g8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1856` (ply 68) [structural distance  $d=110$ , QAP obj=100]: the master played Qxf3; under  $\mu$  our Qd5 plays the role of that game’s Qe3; cross-type roles: our Qd5 as their Rg2, our Bg2 as their Pc4; transferred move d5f3. (d) Retrieved from `ccr1_top_p0003#1238` (ply 77) [structural distance  $d=120$ , QAP obj=98]: the master played Ng3; under  $\mu$  our Ng6 plays the role of that game’s Ne5; cross-type roles: our nd8 as their pc6, our ng5 as their pb2; transferred move g6h4. Synthesis: 1 retrieved analog transferred to Qg8+ (summed QAP objective 76, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



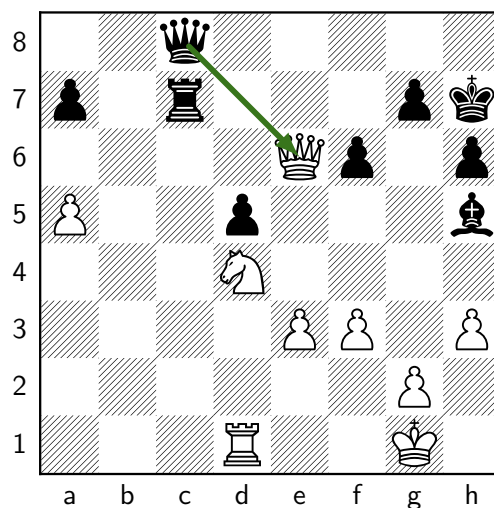
(a)



(b)



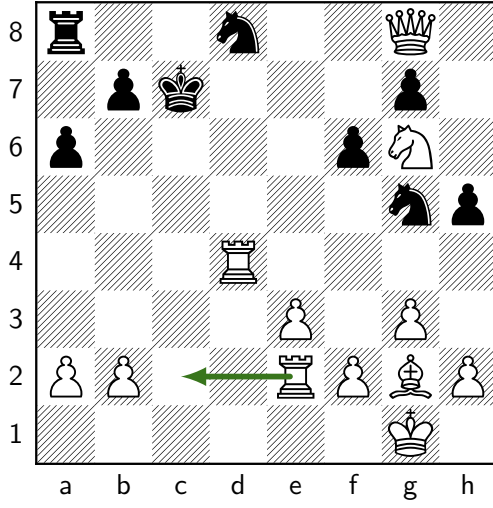
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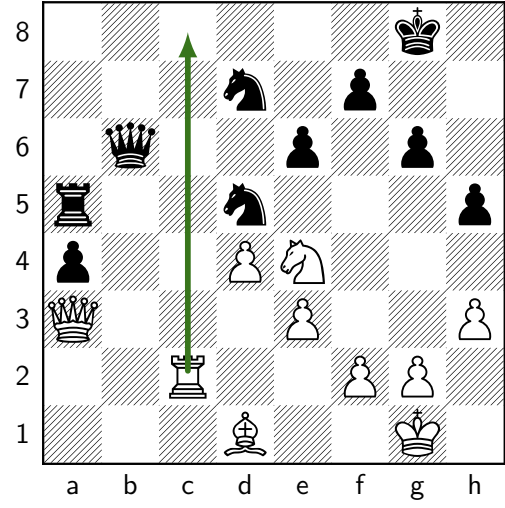
(d)

Figure 29: **Move 35, White: Rxd4+.** (a) Our position; Melanie plays Rxd4+ (d1d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1816` (ply 74) [structural distance  $d=101$ , QAP obj=36]: the master played Qxd8+; under  $\mu$  our *Rd1* plays the role of that game’s *Qb8* (a **cross-type** correspondence); cross-type roles: our *pf6* as their *rd1*, our *Re2* as their *Pe2*; transferred move d1d4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3002` (ply 62) [structural distance  $d=100$ , QAP obj=22]: the master played Rxd5; under  $\mu$  our *Rd1* plays the role of that game’s *Rd1*; cross-type roles: our *Pf2* as their *Ra2*, our *Bg2* as their *Pf2*; transferred move d1d4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#21` (ply 77) [structural distance  $d=88$ , QAP obj=19]: the master played Qxe6; under  $\mu$  our *Rd1* plays the role of that game’s *Qc1* (a **cross-type** correspondence); cross-type roles: our *Qg8* as their *Bh5*, our *ra8* as their *nd4*; transferred move d1d4 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Rxd4+ (summed QAP objective 89, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

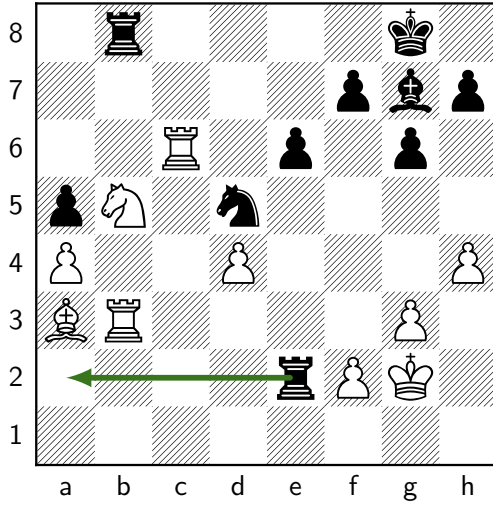




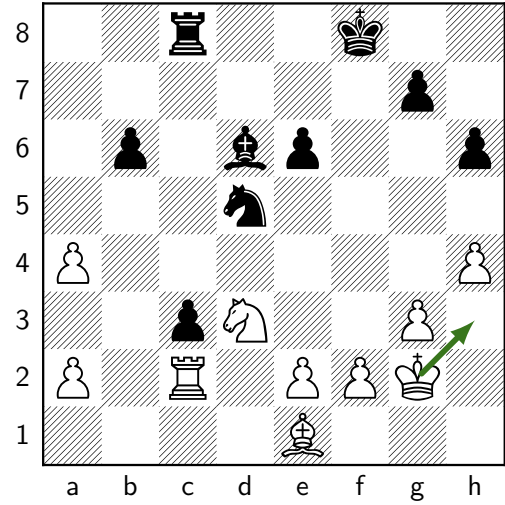
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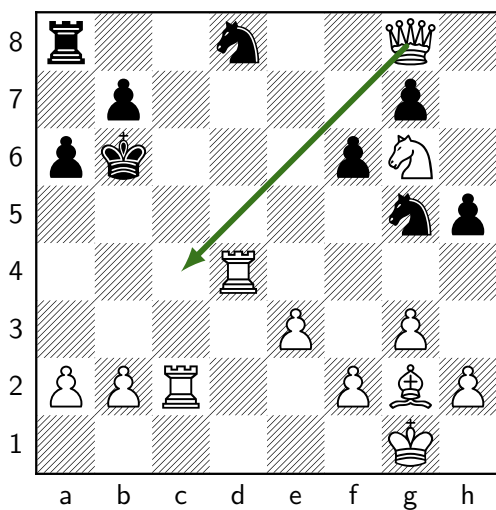


(c)

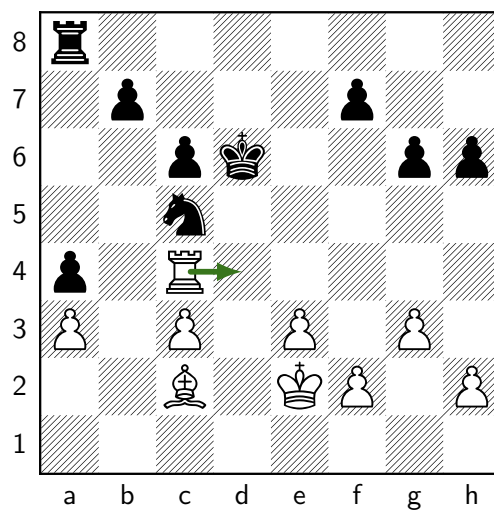


(d)

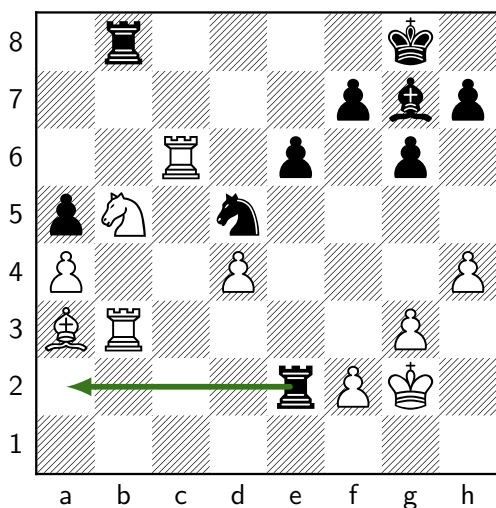
Figure 30: **Move 36, White: Rc2+.** (a) Our position; Melanie plays Rc2+ (e2c2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3081` (ply 78) [structural distance  $d=85$ , QAP obj=59]: the master played Rc8+; under  $\mu$  our *Re2* plays the role of that game’s *Rc2*; cross-type roles: our *ph5* as their *Bd1*, our *Bg2* as their *Pg2*; transferred move **e2c2 (endorses this move)**. (c) Retrieved from `ccr1_top_p0001#2259` (ply 57) [structural distance  $d=91$ , QAP obj=83]: the master played Ra2; under  $\mu$  our *Qg8* plays the role of that game’s *Re7* (a **cross-type** correspondence); cross-type roles: our *Qg8* as their *Rb8*, our *pa6* as their *Bg7*; transferred move **g8g7**. (d) Retrieved from `ccr1_top_p0003#783` (ply 64) [structural distance  $d=91$ , QAP obj=61]: the master played Kh3; under  $\mu$  our *Kg1* plays the role of that game’s *Kg2*; cross-type roles: our *Rd4* as their *Be1*, our *Qg8* as their *Rc2*; transferred move **g1h1**. Synthesis: 1 retrieved analog transferred to Rc2+ (summed QAP objective 59, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



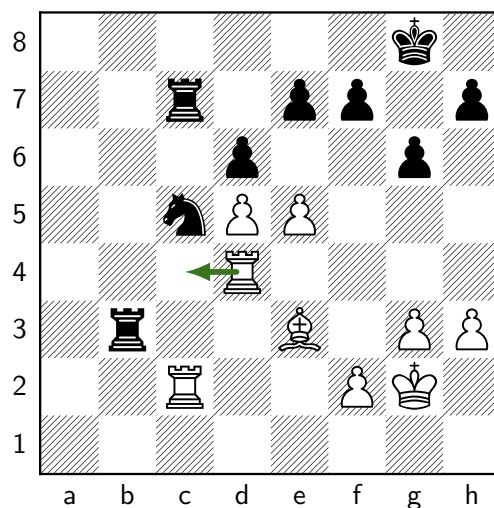
(a)



(b)

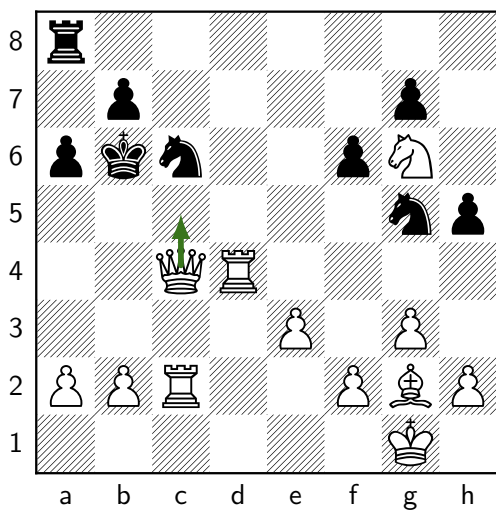


(c)

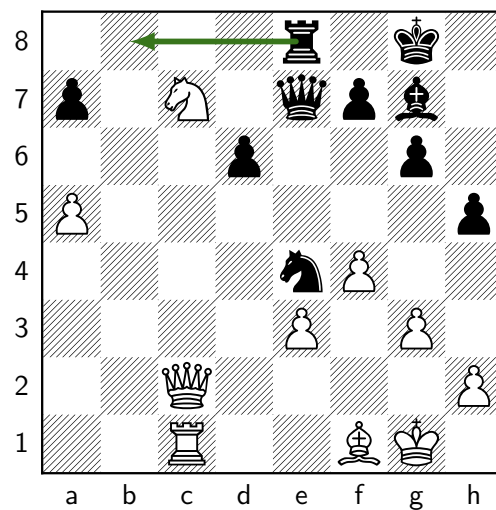


(d)

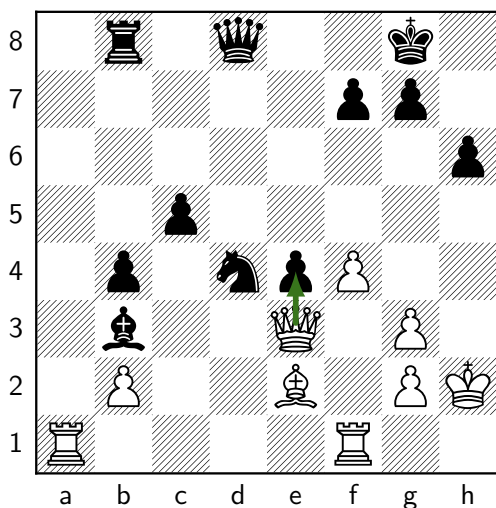
Figure 31: **Move 37, White: Qc4.** (a) Our position; Melanie plays Qc4 (g8c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2712` (ply 70) [structural distance  $d=68$ , QAP obj=15]: the master played Rd4+; under  $\mu$  our Qg8 plays the role of that game’s Rc4 (a **cross-type** correspondence); cross-type roles: our Ng8 as their Bc2, our Bg2 as their Pa3; transferred move g8c4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2259` (ply 57) [structural distance  $d=75$ , QAP obj=81]: the master played Ra2; under  $\mu$  our Qg8 plays the role of that game’s Re7 (a **cross-type** correspondence); cross-type roles: our Qg8 as their Rb8, our nd8 as their Pe6; transferred move g8g7. (d) Retrieved from `ccr1_top_p0002#3971` (ply 68) [structural distance  $d=77$ , QAP obj=70]: the master played Rdc4; under  $\mu$  our Rc2 plays the role of that game’s Rd4; cross-type roles: our Qg8 as their Rc2, our Qg8 as their rb3; transferred move c2c3. Synthesis: 2 retrieved analogs transferred to Qc4 (summed QAP objective 31, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



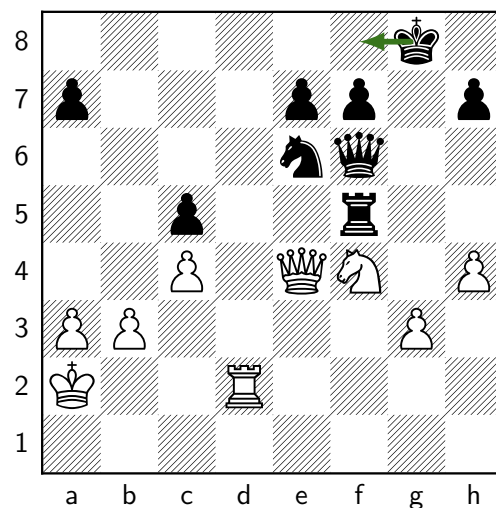
(a)



(b)

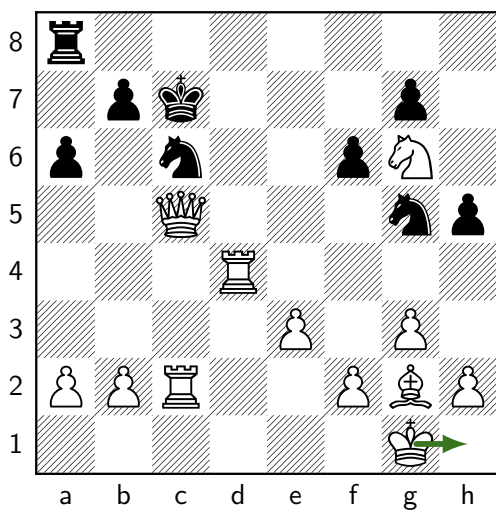


(c)

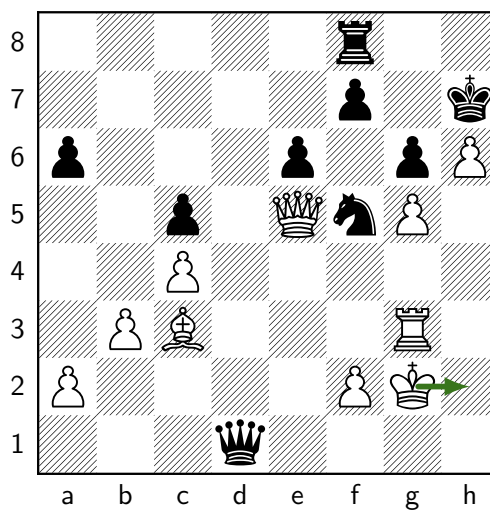


(d)

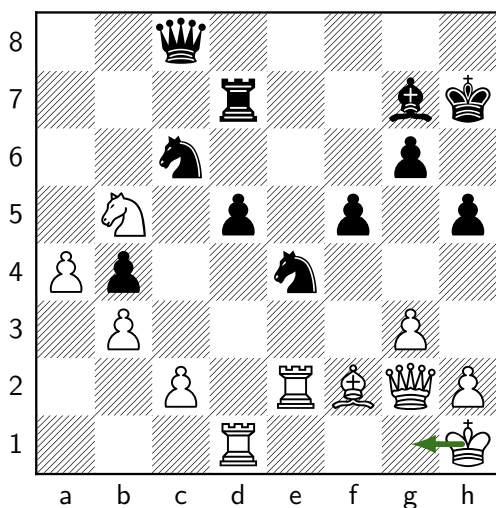
Figure 32: **Move 38, White: Qc5+.** (a) Our position; Melanie plays Qc5+ (c4c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2125` (ply 79) [structural distance  $d=106$ , QAP obj=57]: the master played Rb8; under  $\mu$  our *Rd4* plays the role of that game’s *Re1*; cross-type roles: our *Bg2* as their *Pa7*, our *pb7* as their *Bg7*; transferred move d4d1. (c) Retrieved from `ccr1_top_p0000#3258` (ply 58) [structural distance  $d=108$ , QAP obj=42]: the master played Qxe4; under  $\mu$  our *Qc4* plays the role of that game’s *Qe3*; cross-type roles: our *Pg3* as their *Ra1*, our *ng5* as their *Rf1*; transferred move c4a4. (d) Retrieved from `ccr1_top_p0003#1601` (ply 83) [structural distance  $d=107$ , QAP obj=40]: the master played Kf8; under  $\mu$  our *Kg1* plays the role of that game’s *Kg1*; cross-type roles: our *ra8* as their *Pf7*, our *pa6* as their *Qf6*; transferred move g1h1. Synthesis: no retrieved analog transferred to Qc5+ – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



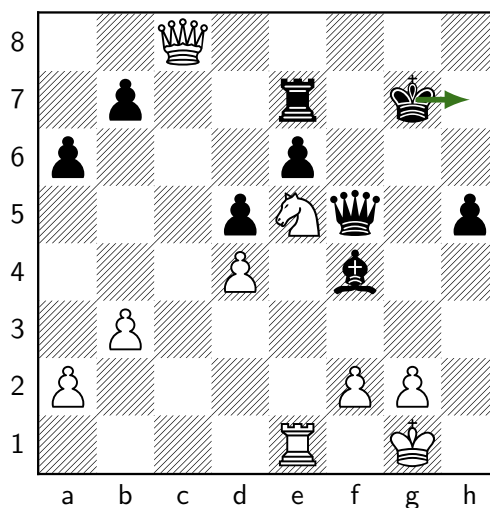
(a)



(b)



(c)



(d)

Figure 33: **Move 39, White: Kh1.** (a) Our position; Melanie plays Kh1 ( $g1h1$ ). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2260` (ply 74) [structural distance  $d=111$ , QAP obj=56]: the master played Kh2; under  $\mu$  our  $Kg1$  plays the role of that game’s  $Kg2$ ; cross-type roles: our  $Pg3$  as their  $qd1$ , our  $Ng6$  as their  $Pb3$ ; transferred move  $g1h1$  (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1096` (ply 80) [structural distance  $d=110$ , QAP obj=50]: the master played  $Kg1$ ; under  $\mu$  our  $Kg1$  plays the role of that game’s  $Kh1$ ; cross-type roles: our  $Pb2$  as their  $Rd1$ , our  $Bg2$  as their  $Pc2$ ; transferred move  $g1h1$  (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3756` (ply 77) [structural distance  $d=111$ , QAP obj=40]: the master played Kh7; under  $\mu$  our  $Kg1$  plays the role of that game’s  $Kg2$ ; cross-type roles: our  $nc6$  as their  $qc8$ , our  $Rd4$  as their  $Pb7$ ; transferred move  $g1h1$  (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Kh1 (summed QAP objective 226, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

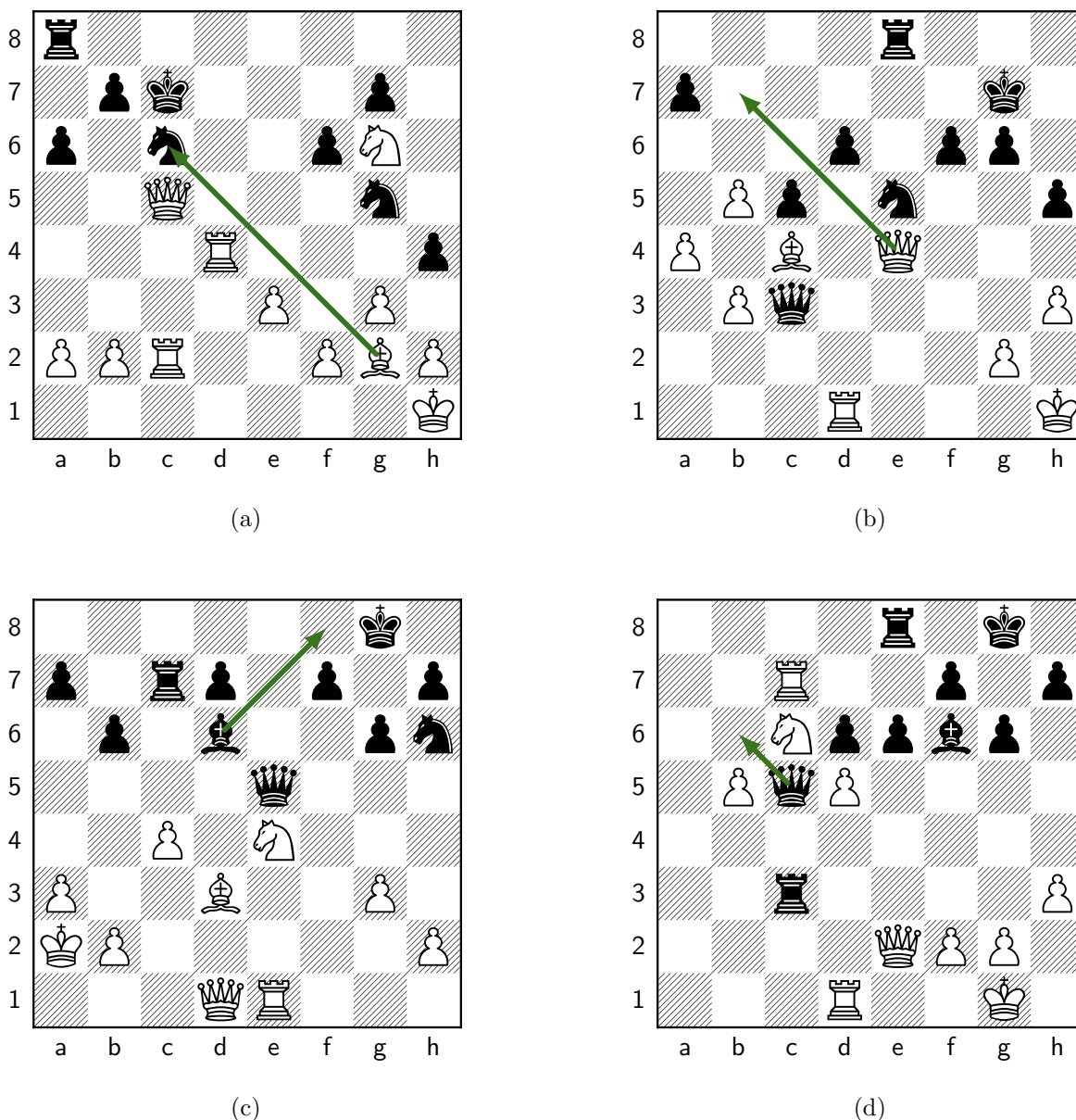
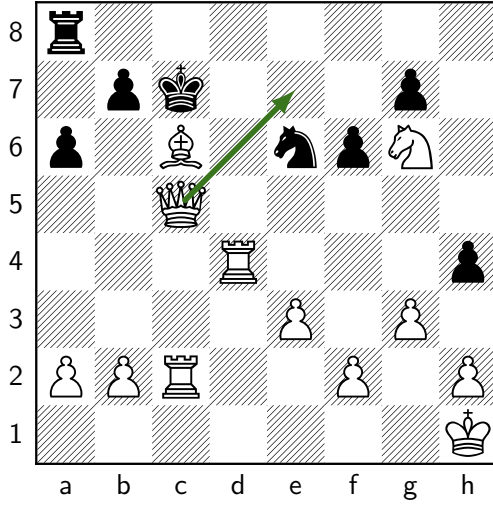
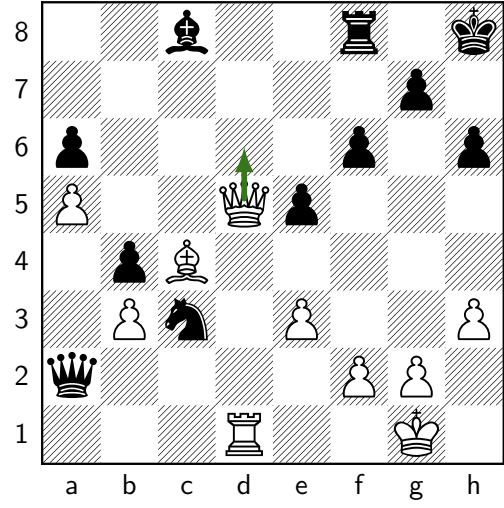


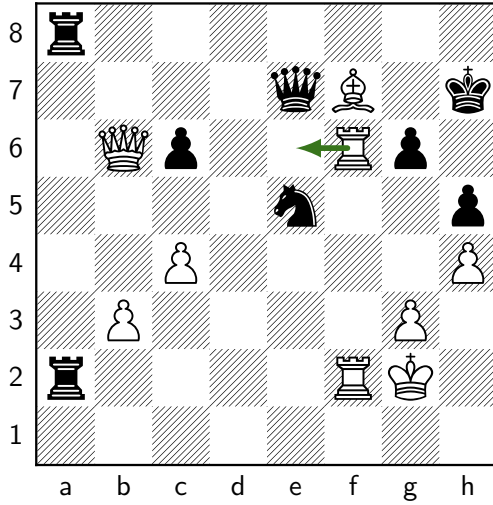
Figure 34: **Move 40, White: Bxc6.** (a) Our position; Melanie plays Bxc6 (g2c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1470` (ply 80) [structural distance  $d=96$ , QAP obj=83]: the master played Qb7+; under  $\mu$  our Qc5 plays the role of that game’s Qe4; cross-type roles: our Bg2 as their Rd1, our Rd4 as their Pb3; transferred move c5e7. (c) Retrieved from `ccr1_top_p0003#1744` (ply 77) [structural distance  $d=105$ , QAP obj=77]: the master played Bf8; under  $\mu$  our Rd4 plays the role of that game’s Bd3 (a **cross-type** correspondence); cross-type roles: our pg7 as their Rc7, our Ng6 as their Pd7; transferred move d4a4. (d) Retrieved from `ccr1_top_p0003#1469` (ply 67) [structural distance  $d=106$ , QAP obj=73]: the master played Qb6; under  $\mu$  our Qc5 plays the role of that game’s Qc4; cross-type roles: our Bg2 as their Re8, our pb7 as their rc7; transferred move c5a5. Synthesis: no retrieved analog transferred to Bxc6 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



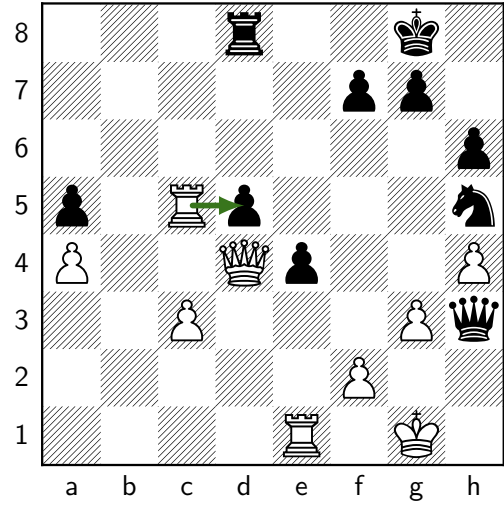
(a)



(b)

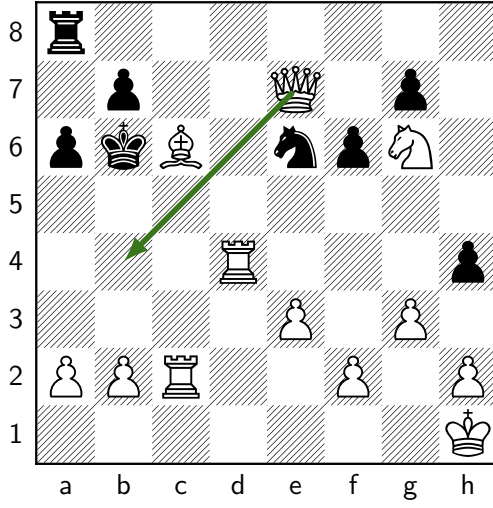


(c)

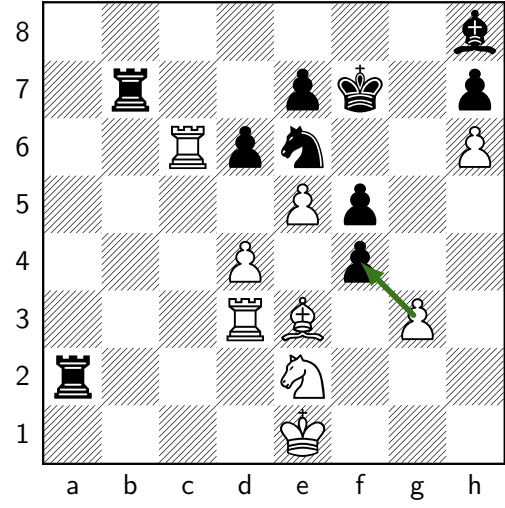


(d)

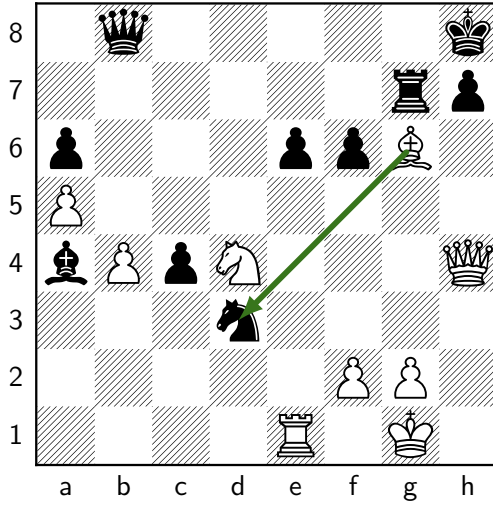
Figure 35: **Move 41, White: Qe7+.** (a) Our position; Melanie plays Qe7+ (c5e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#833` (ply 68) [structural distance  $d=104$ , QAP obj=60]: the master played Qd6; under  $\mu$  our Qc5 plays the role of that game’s Qd5; cross-type roles: our Bc6 as their qa2, our Ng6 as their pb4; transferred move c5e7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#850` (ply 96) [structural distance  $d=104$ , QAP obj=57]: the master played Re6; under  $\mu$  our Qc5 plays the role of that game’s Rf6 (a **cross-type** correspondence); cross-type roles: our Bc6 as their ra2, our pb7 as their Rf2; transferred move c5a7. (d) Retrieved from `ccr1_top_p0001#2200` (ply 70) [structural distance  $d=99$ , QAP obj=56]: the master played Rxd5; under  $\mu$  our Bc6 plays the role of that game’s Rc5 (a **cross-type** correspondence); cross-type roles: our Pg3 as their Re1, our Ng6 as their Pf2; transferred move c6d5. Synthesis: 1 retrieved analog transferred to Qe7+ (summed QAP objective 60, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



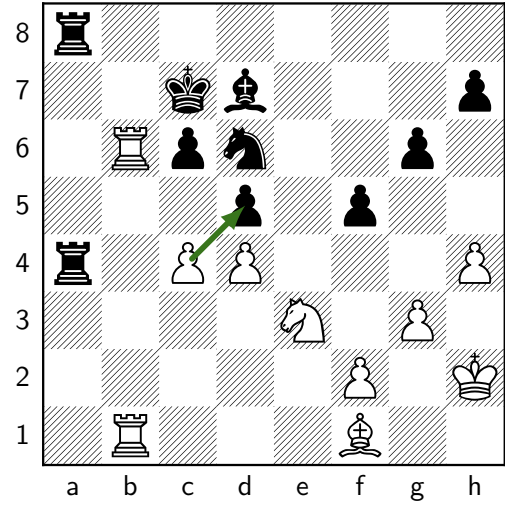
(a)



(b)



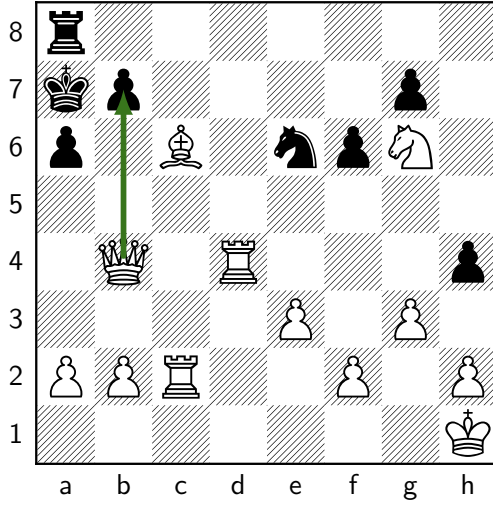
(c)



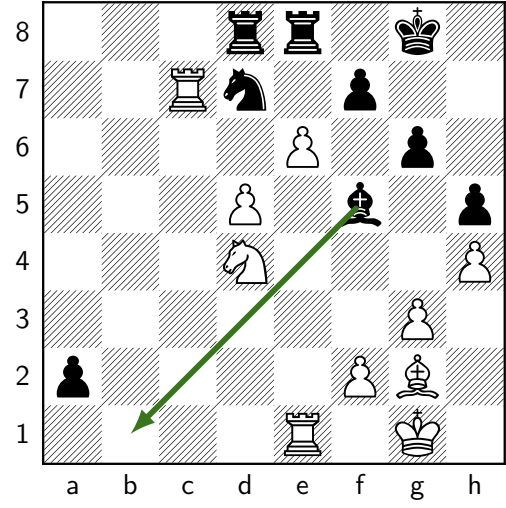
(d)

Figure 36: **Move 42, White: Qb4+.** (a) Our position; Melanie plays Qb4+ (e7b4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3856` (ply 68) [structural distance  $d=117$ , QAP obj=76]: the master played `gxf4`; under  $\mu$  our `Qe7` plays the role of that game’s `Pg3` (a **cross-type** correspondence); cross-type roles: our `Pa2` as their `ra2`, our `Qe7` as their `Ne2`; transferred move `e7b4` (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#850` (ply 70) [structural distance  $d=120$ , QAP obj=78]: the master played `Bxd3`; under  $\mu$  our `Bc6` plays the role of that game’s `Bg6`; cross-type roles: our `ne6` as their `Re1`, our `Qe7` as their `nd3`; transferred move `c6f3`. (d) Retrieved from `ccr1_top_p0003#1807` (ply 78) [structural distance  $d=122$ , QAP obj=69]: the master played `cxd5`; under  $\mu$  our `Pf2` plays the role of that game’s `Pc4`; cross-type roles: our `Ph2` as their `Bf1`, our `Ng6` as their `Pf2`; transferred move `f2f3`. Synthesis: 1 retrieved analog transferred to Qb4+ (summed QAP objective 76, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

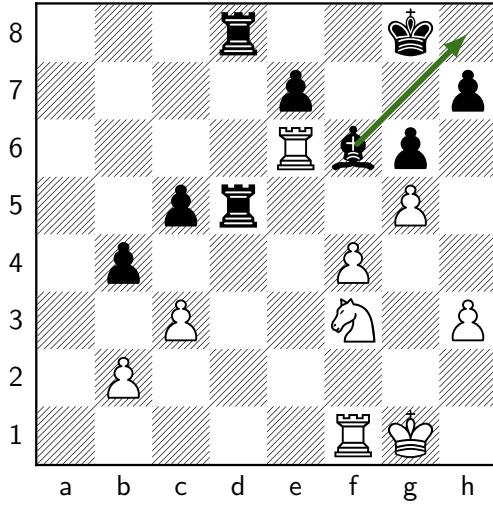




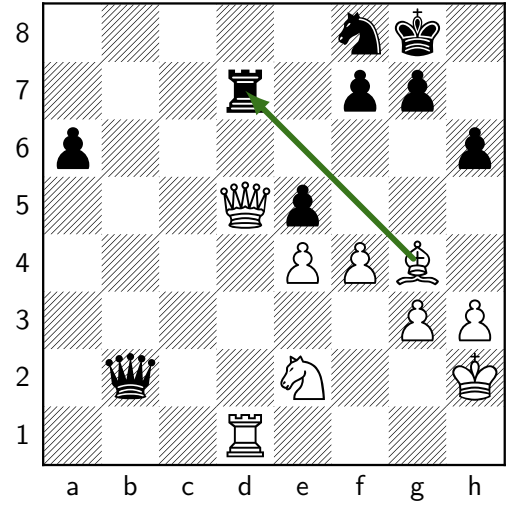
(a)



(b)



(c)



(d)

Figure 37: **Move 43, White: Qxb7#.** (a) Our position; Melanie plays Qxb7# (b4b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1036` (ply 79) [structural distance  $d=96$ , QAP obj=71]: the master played Bb1; under  $\mu$  our *Rd4* plays the role of that game’s *Bf4* (a **cross-type** correspondence); cross-type roles: our *Qb4* as their *Re8*, our *pb7* as their *rc7*; transferred move d4d7. (c) Retrieved from `ccr1_top_p0002#279` (ply 75) [structural distance  $d=89$ , QAP obj=68]: the master played Bh8; under  $\mu$  our *Bc6* plays the role of that game’s *Bf3*; cross-type roles: our *Bc6* as their *Pe7*, our *Rc2* as their *Ph7*; transferred move c6g2. (d) Retrieved from `ccr1_top_p0000#475` (ply 74) [structural distance  $d=97$ , QAP obj=49]: the master played Bxd7; under  $\mu$  our *Pe3* plays the role of that game’s *Bg4* (a **cross-type** correspondence); cross-type roles: our *Bc6* as their *qb2*, our *pb7* as their *Ne2*; transferred move e3e4. Synthesis: no retrieved analog transferred to Qxb7# – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.